第1章 绪论

第1章 绪论

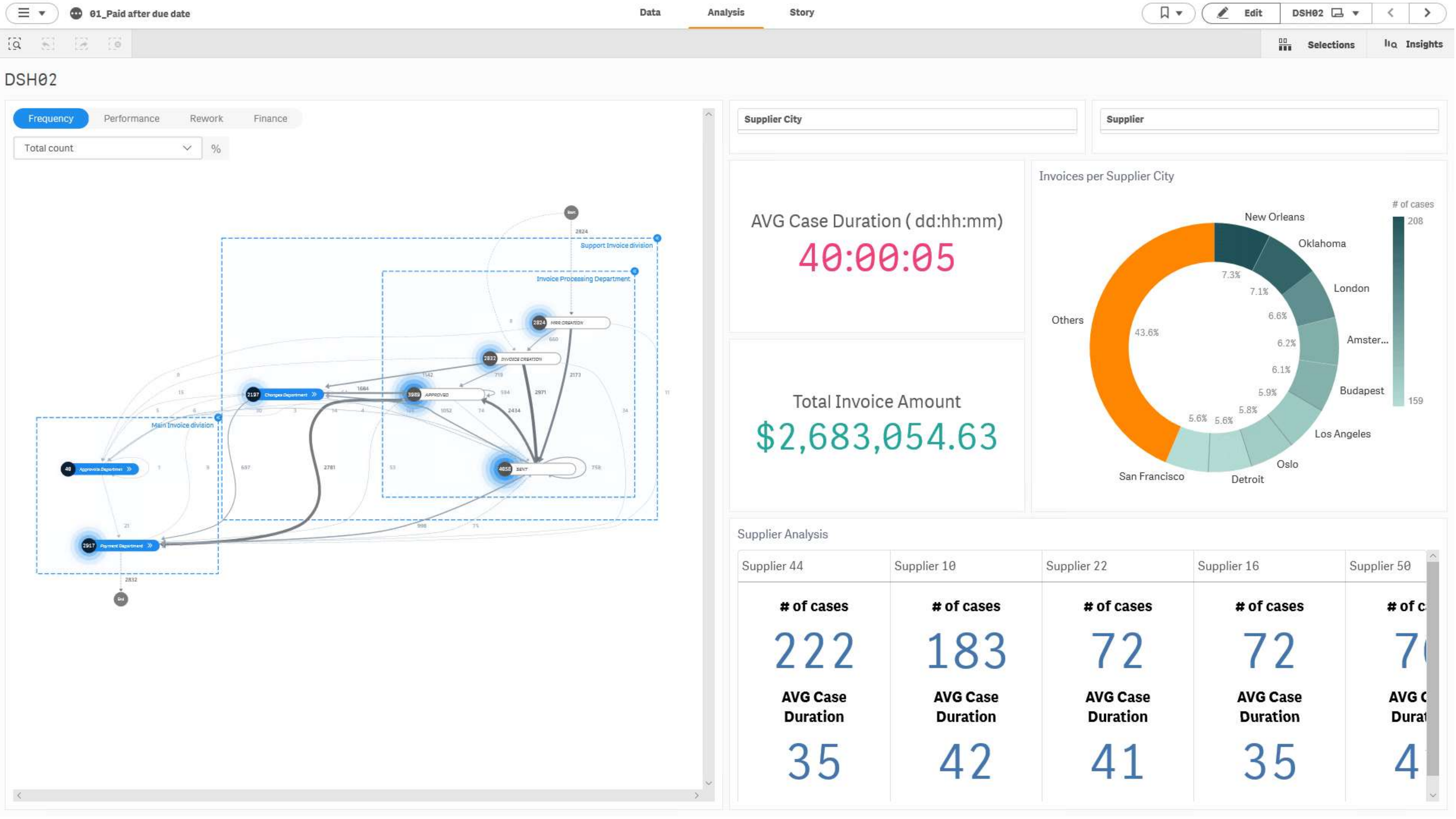
数据管理

数据库系统

DBMS 的特色

■ 数据管理

- 何为数据？何为管理？
- 数据管理有哪些操作？
- 数据管理有哪些方案？



01_Paid after due date
Data
Analysis
Story
Edit
DSH02
Selections
Insights
DSH02
Frequency
Performance
Rework
Finance
Total count
%
Support Invoice division
Invoice Processing Department
Main Invoice division
Supplier City
Supplier
AVG Case Duration (dd:hh:mm)
40:00:05
Total Invoice Amount
$2,683,054.63
Invoices per Supplier City
# of cases
208
159
New Orleans
Oklahoma
London
Amster...
Budapest
Los Angeles
Oslo
Detroit
San Francisco
Others
7.3%
7.1%
6.6%
6.2%
6.1%
5.9%
5.8%
5.6%
5.6%
43.6%
Supplier Analysis
Supplier 44
# of cases
222
AVG Case Duration
35
Supplier 10
# of cases
183
AVG Case Duration
42
Supplier 22
# of cases
72
AVG Case Duration
41
Supplier 16
# of cases
72
AVG Case Duration
35
Supplier 50

设计时

程序 = 数据结构 + 算法

运行时

进程　数据　指令

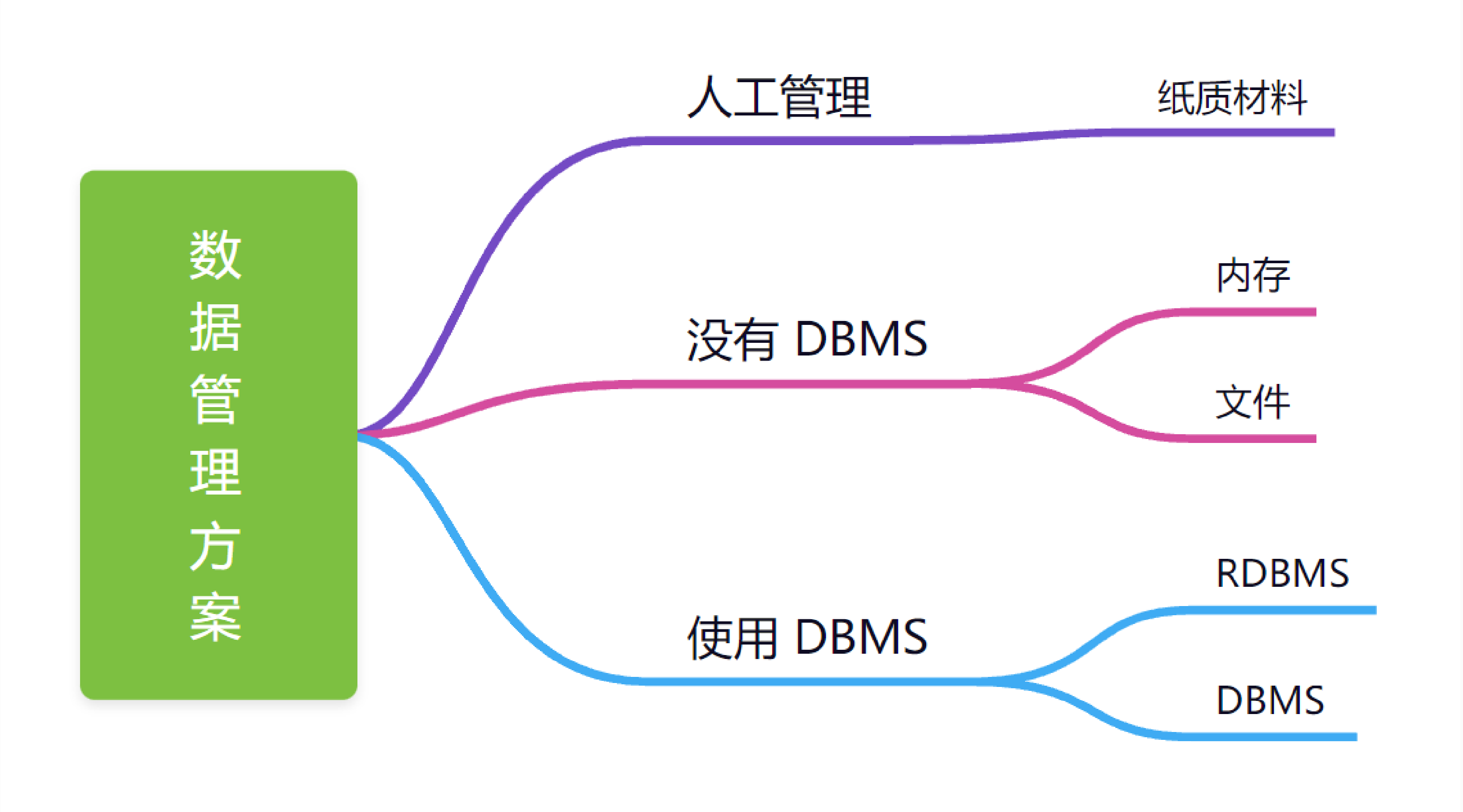
何为数据?

数据是**能够被记录**且**具有实际含义**的**已知事实**。

- 大数据
- 元数据

create table *instructor*
(*ID* **varchar** (5),
name **varchar** (20) **not null**,
dept_name **varchar** (20),
salary **numeric** (8,2),
primary key (*ID*),
foreign key (*dept_name*) **references** *department*);

- 存数据 W
  - 增加 / 删除数据
  - 修改数据
- 取数据 R
  - 条件查询
  - 聚合查询
- 其他 O
  - 安全性约束
  - 完整性约束
  - 数据共享
  - 并发控制
  - …



DBMS：Database Management System 数据库管理系统


DEMO

```
~/2nd/db-demo/01-introduction (master) » ls                    wangding@amtb
scores.csv  score.sh
~/2nd/db-demo/01-introduction (master) »                       wangding@amtb
```

```
~/2nd/db-demo/01-introduction (master) » tail -10 scores.csv
"14406524","顾潘婷","10","30","20"
"1345327","杨翼铭","100","50","60"
"1342624","卢祎萌","90","30","50"
"1345401","安海铭","100","70","90"
"14405629","袁霞","10","90","70"
"14407704","周耘如","80","70","10"
"14407705","吴丽萍","100","30","60"
"14403310","徐子悦","90","100","70"
"14404725","陈伊婷","60","80","70"
"14401215","黄麒","20","80","30"
~/2nd/db-demo/01-introduction (master) »
```

```
# 查看前十行数据
head scores.csv

# 查找学号为 14407503 的一行信息
grep 14407503 scores.csv

# 查看 C 语言成绩
head scores.csv | awk -F ',' '{print $3}'

# 去掉每列的双引号
head scores.csv | sed 's/\"//g'
```

```
# 交互式读取用户输入
read col

# 查看变量
echo $col

# 查看 C 语言成绩
head scores.csv | awk -F ',' '{print $3}'

# 查看 col 变量指定的列
head scores.csv | awk -F ',' -v n=$col '{print $n}'
```

- 基于文件的数据管理方案的缺点
  - 当文件格式发生变化，要修改应用程序
  - 文件修改可能造成数据不一致，破坏数据正确性
  - 没有索引，数据查找效率低
  - 只能对整个文件进行访问控制，数据安全性差
  - 没有并发控制，多个应用程序同时读写文件可能产生冲突

	Filesystem	Database
DEFINITION	A process that manages how and where data on a storage disk is stored, accessed and managed	An organized collection of data that can be easily storage accessed, managed and updated
DATA CONSISTENCY	Has high data inconsistency	Maintains data consistency
STRUCTURE	Structure is simple	Structure is complex
DATA SHARING	Data sharing is hard	Data sharing is easy
REDUNDANCY	There is high redundancy	There is low redundancy
SECURITY	Not very secure	More secure
BACKUP AND RECOVERY	No backup and recovery process	There is backup recovery

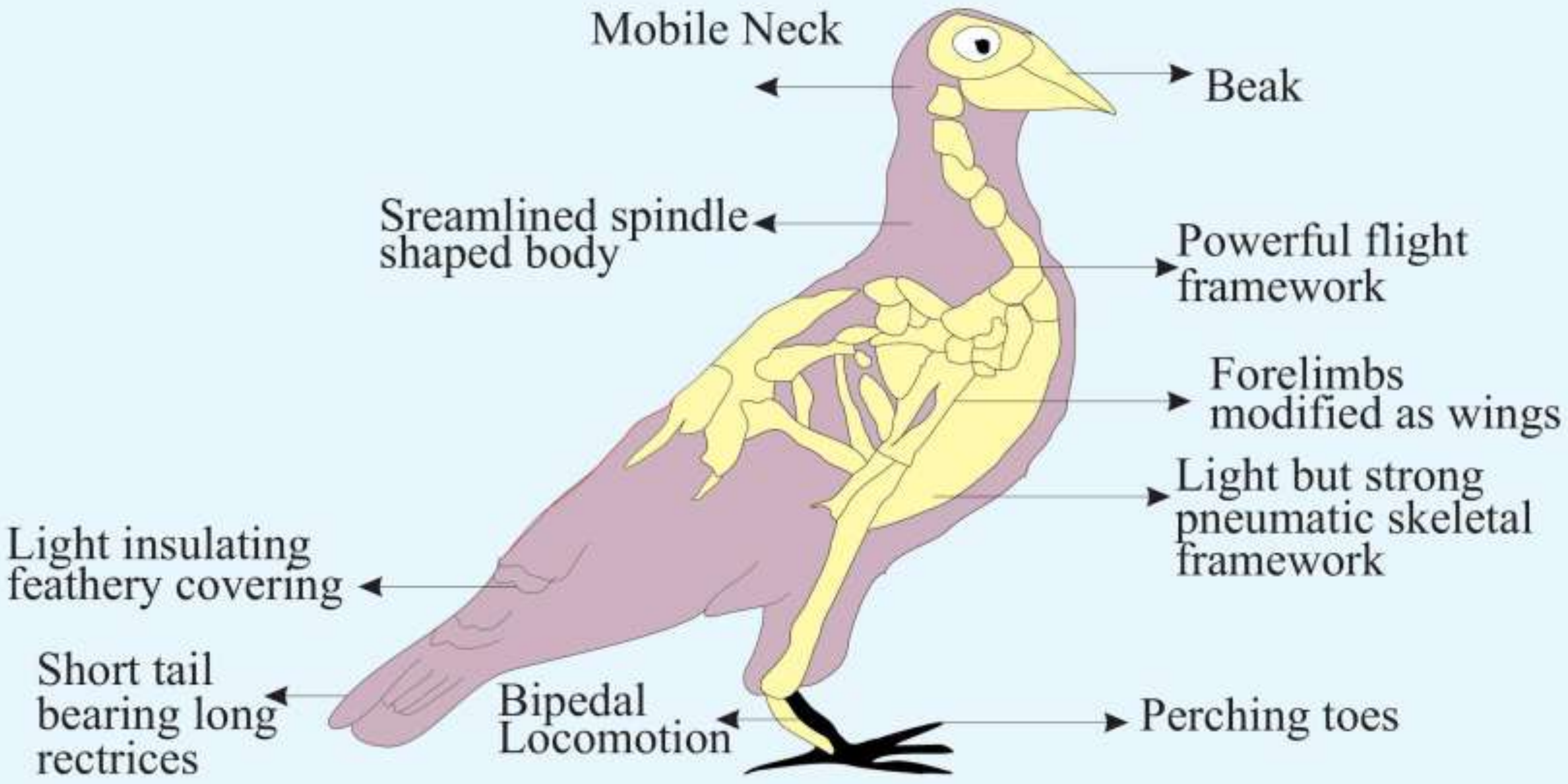
数据管理

数据库系统

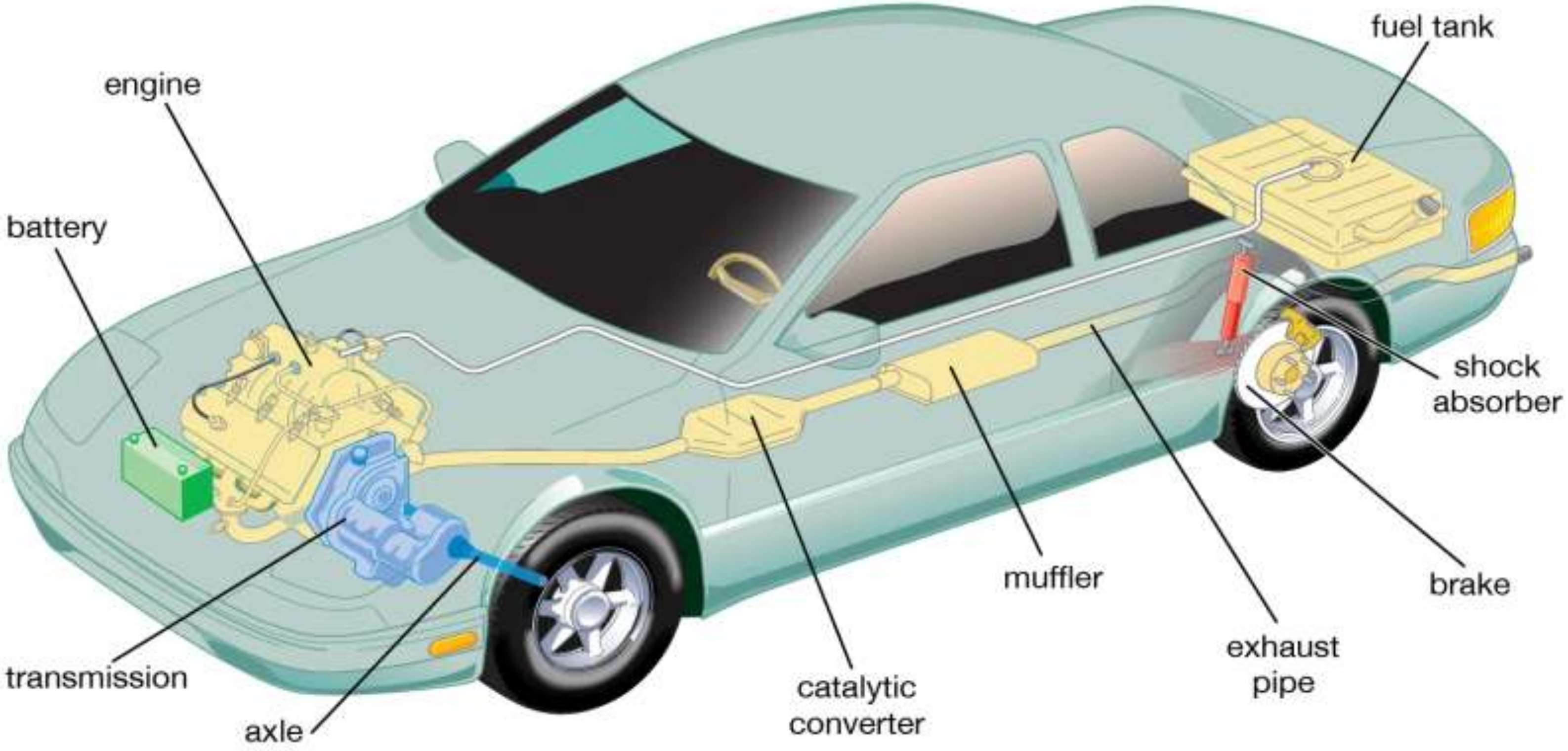
DBMS 的特色

■ 数据库（Database）

■ 系统（System）

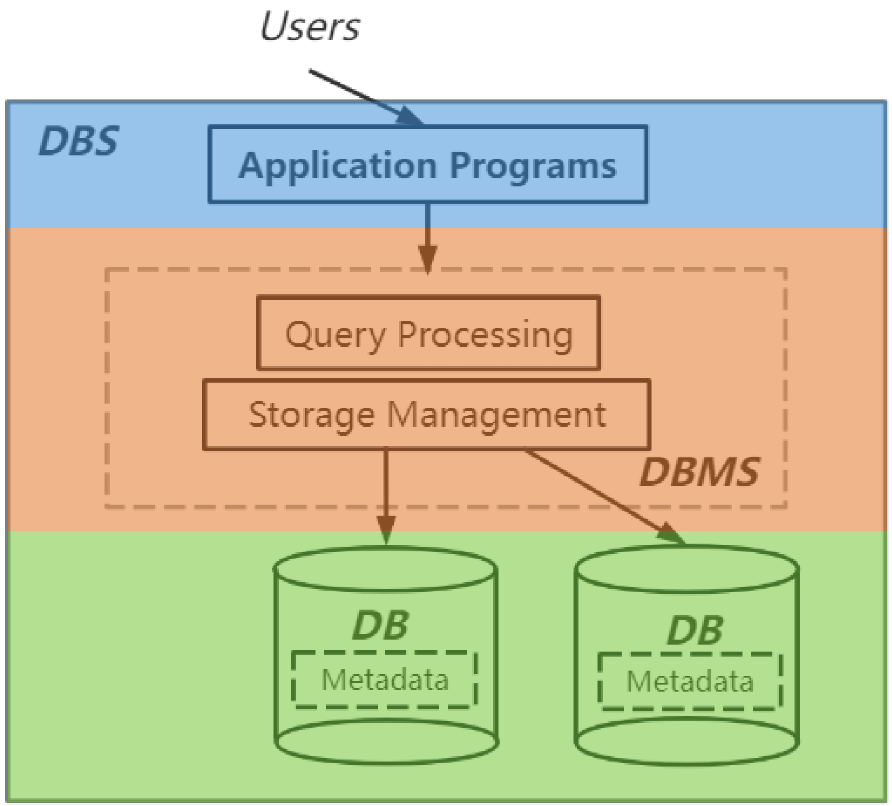

Mobile Neck
Beak
Sreamlined spindle shaped body
Powerful flight framework
Forelimbs modified as wings
Light but strong pneumatic skeletal framework
Light insulating feathery covering
Short tail bearing long rectrices
Bipedal Locomotion
Perching toes

■ 系统（System）



fuel tank
engine
battery
shock
absorber
muffler
brake
exhaust
pipe
transmission
catalytic
converter
axle

数据库系统（DBS：Database System）



数据库系统是高效**组织**和**管理**数据的系统，由**数据库**、**数据库管理系统**和**应用程序**三部分构成。

■ **应用程序**（Application programs）

应用程序提供了用户与数据库系统的**交互界面**。

■ **数据库管理系统**（DBMS：Database Management System）

DBMS is **a general-purpose system software** that faciliates the organization, storage, manipulation, control, and maintainence of databases among various users and application programs.

■ **数据库**（Database）

Database is **a set of related data** that is organized, shared, and persistent.

SQLite

ORACLE

PostgreSQL

Microsoft SQL Server

dbisam

MySQL

Microsoft SQL Azure

elevatedb

TURBODB Embedded

CouchDB
relax

Redis

Neo4j
the graph database

Cassandra

mongoDB

membase

riak

DBMS 分类

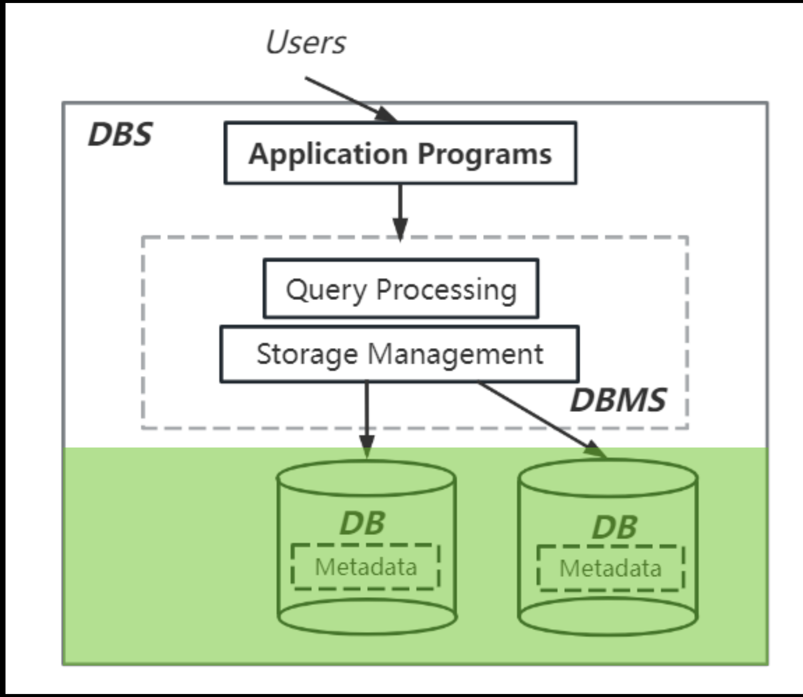
- 按逻辑数据模型分
  - RDBMS
  - DBMS
- 按数据库规模分
  - 网络/企业数据库
  - 桌面数据库
- 按是否收费分
  - 商业收费版
  - 开源免费版


DEMO

```
~ » mysql -e 'show databases;'                                                    wangding@OFFICE
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| northwind          |
| performance_schema |
| sys                |
+--------------------+
```

```
~ » sudo ls /var/lib/mysql/ --color                                               wangding@OFFICE
aria_log.00000001  ca-key.pem       client-key.pem  ib_logfile0  mysql            northwind           public_key.pem   sys
aria_log_control   ca.pem           ib_buffer_pool  ib_logfile1  mysql.sock       performance_schema  server-cert.pem
auto.cnf           client-cert.pem  ibdata1         ibtmp1       mysql.sock.lock  private_key.pem     server-key.pem
```

```
~ »                                                                               wangding@OFFICE
```



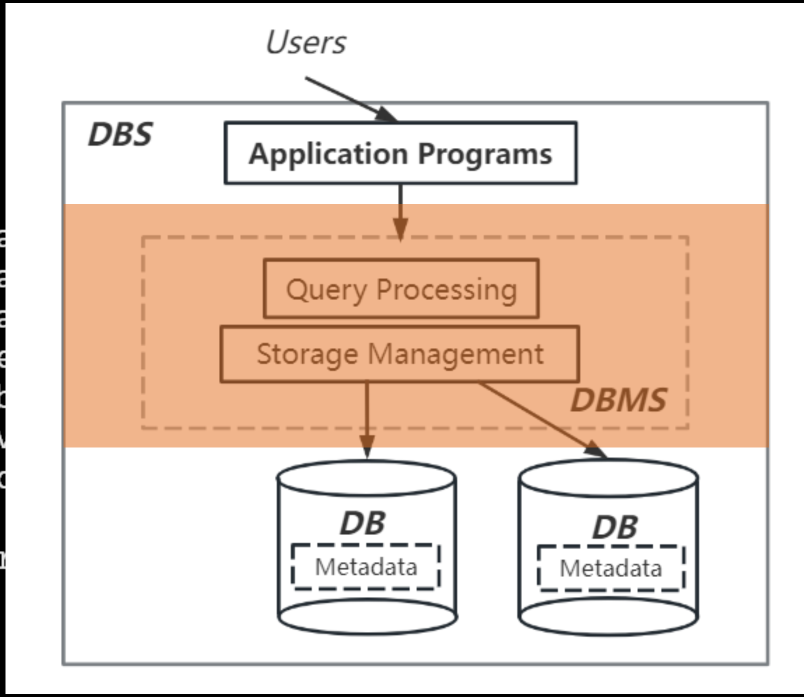
注意：

- 你查看到的信息可能跟本图不同
- mysql, sys, performance_schema 一定相同
- 一个文件夹，如 northwind 对应左图的一个圆柱

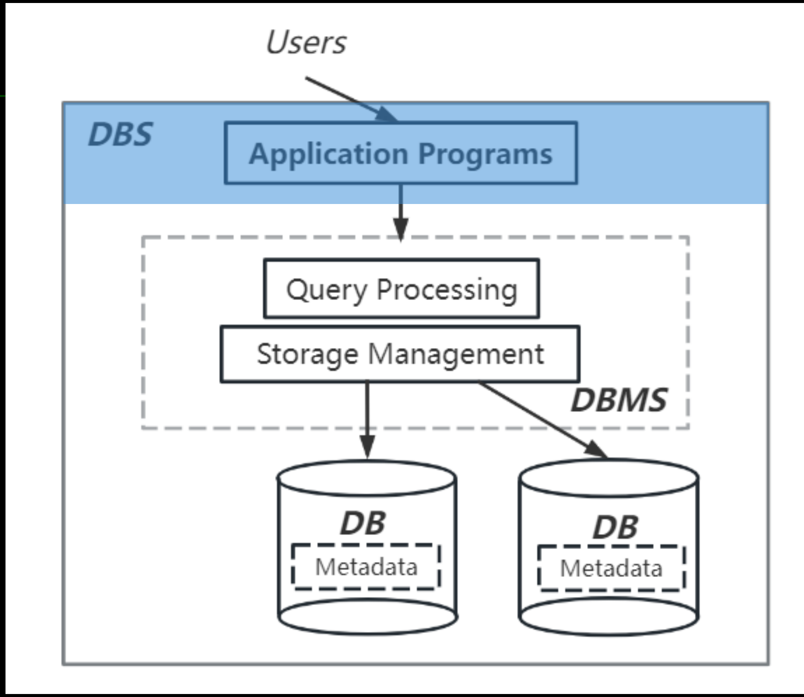
```
~ » mysql -e 'show tables from northwind;'                    wangding@OFFICE
+----------------------+
| Tables_in_northwind |
+----------------------+
| countries            |
| departments          |
| employees            |
| job_grades           |
| job_history          |
| jobs                 |
| locations            |
| order                |
| regions              |
+----------------------+
~ » sudo ls /var/lib/mysql/northwind --color                  wangding@OFFICE
countries.frm  departments.frm  employees.ibd   job_history.frm  jobs.ibd       order.frm    regions.ibd
countries.ibd  departments.ibd  job_grades.frm  job_history.ibd  locations.frm  order.ibd
db.opt         employees.frm    job_grades.ibd  jobs.frm         locations.ibd  regions.frm
~ »                                                           wangding@OFFICE
```

```
~ » mysql -e 'show tables from northwind;'                                    wangding@amtb
+----------------------+
| Tables_in_northwind |
+----------------------+
| countries            |
| departments          |
| employees            |
| job_grades           |
| job_history          |
| jobs                 |
| locations            |
| order                |
| regions              |
+----------------------+
----------------------------------------------------------------------------------------------
~ » sudo ls /var/lib/mysql/northwind --color                                  wangding@amtb
countries.ibd    employees.ibd   job_history.ibd  locations.ibd  regions.ibd
departments.ibd  job_grades.ibd  jobs.ibd         order.ibd
----------------------------------------------------------------------------------------------
~ »                                                                           wangding@amtb
```

```
~/doc/log(master) » ps aux | grep mysql                                                          wangding@OFFICE
mysql      1346  0.1 10.4 1251380 195044 ?     Sl   09:14   0:36 /usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid
wangding   2335  0.0  0.0 112820   992 pts/0    S+   16:15   0:00 grep --color=auto --exclude-dir=.bzr --exclude-dir=CVS --exclude-dir=.
git --exclude-dir=.hg --exclude-dir=.svn mysql

~/doc/log(master) » systemctl status mysqld                                                      wangding@OFFICE
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; vendor preset: disabled)
   Active: active (running) since 三 2023-02-08 09:14:46 CST; 7h ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
  Process: 1316 ExecStart=/usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid $MYSQLD_OPTS (code=exited, status=0/SUCCESS
)
  Process: 892 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
 Main PID: 1346 (mysqld)
   Memory: 277.0M
   CGroup: /system.slice/mysqld.service
           └─1346 /usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid

2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.109910Z 0 [Note] Skipping genera                                  ory.
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.112445Z 0 [Warning] CA certifica
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.112504Z 0 [Note] Skipping genera                                  ory.
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.118057Z 0 [Note] Server hostname
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.118310Z 0 [Note] IPv6 is availab
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.118321Z 0 [Note]   - '::' resolv
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.118338Z 0 [Note] Server socket c
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.317928Z 0 [Note] InnoDB: Buffer
2月 08 09:14:46 OFFICE mysqld[1316]: 2023-02-08T01:14:46.501509Z 0 [Note] Event Scheduler
2月 08 09:14:46 OFFICE systemd[1]: Started MySQL Server.
Hint: Some lines were ellipsized, use -l to show in full.

~/doc/log(master) »                                                                              wangding@OFFICE
```

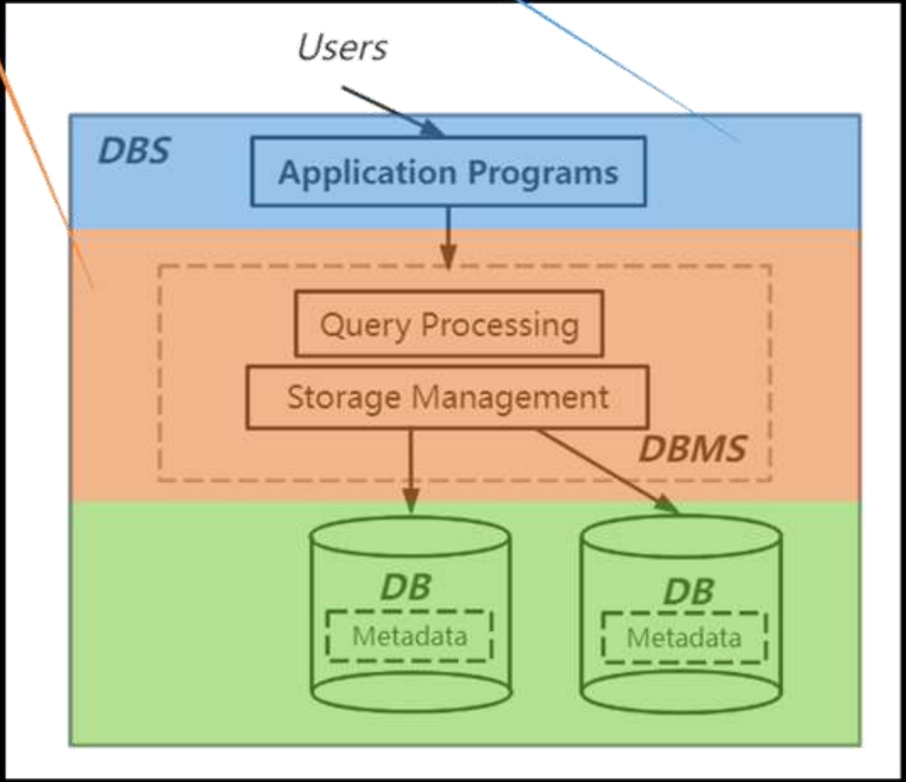


```
~ » ps aux | grep mysql                                                                    wangding@OFFICE
mysql       1346  0.1 10.4 1317180 195044 ?       Sl   09:14   0:37 /usr/sbin/mysqld --daemonize --pid-file=/var/run/mysqld/mysqld.pid
wangding    3706  0.0  0.2 135708  4044 pts/3    S+   16:27   0:00 mysql
wangding    3738  0.0  0.2 135828  4152 pts/1    S+   16:28   0:00 mysql
wangding    3785  0.0  0.0 112820   992 pts/2    S+   16:33   0:00 grep --color=auto --exclude-dir=.bzr --exclude-dir=CVS --exclude-dir=.
git --exclude-dir=.hg --exclude-dir=.svn mysql
------------------------------------------------------------
~ » which mysql                                                                            wangding@OFFICE
/usr/bin/mysql
------------------------------------------------------------
~ »                                                                                        wangding@OFFICE

Connection id:          10
Current database:
Current user:           wangding@localhost
SSL:                    Cipher in use is ECDHE-RSA-AES128-GCM-SHA256
Current pager:          less
Using outfile:          ''
Using delimiter:        ;
Server version:         5.7.30 MySQL Community Server (GPL)
Protocol version:       10
Connection:             lab via TCP/IP
Server characterset:    utf8
Db     characterset:    utf8
Client characterset:    utf8
Conn.  characterset:    utf8
TCP port:               3306
Uptime:                 7 hours 19 min 23 sec

Threads: 2  Questions: 49  Slow queries: 0  Opens: 108  Flush tables: 1  Open tables: 101  Queries per second avg: 0.001
--------------

(wangding@lab) [(none)]>
[0] 0:mysql*                                                                      "mysql" 16:34 08-Feb-23
```



```
~ » yum list installed |grep mysql
Repodata is over 2 weeks old. Install yum-cron? Or run: yum makecache fast
mysql-community-client.x86_64        5.7.30-1.el7          @mysql57-community
mysql-community-common.x86_64        5.7.30-1.el7          @mysql57-community
mysql-community-libs.x86_64          5.7.30-1.el7          @mysql57-community
mysql-community-libs-compat.x86_64   5.7.30-1.el7          @mysql57-community
mysql-community-server.x86_64        5.7.30-1.el7          @mysql57-community
mysql57-community-release.noarch     el7-10                installed
~ »
```

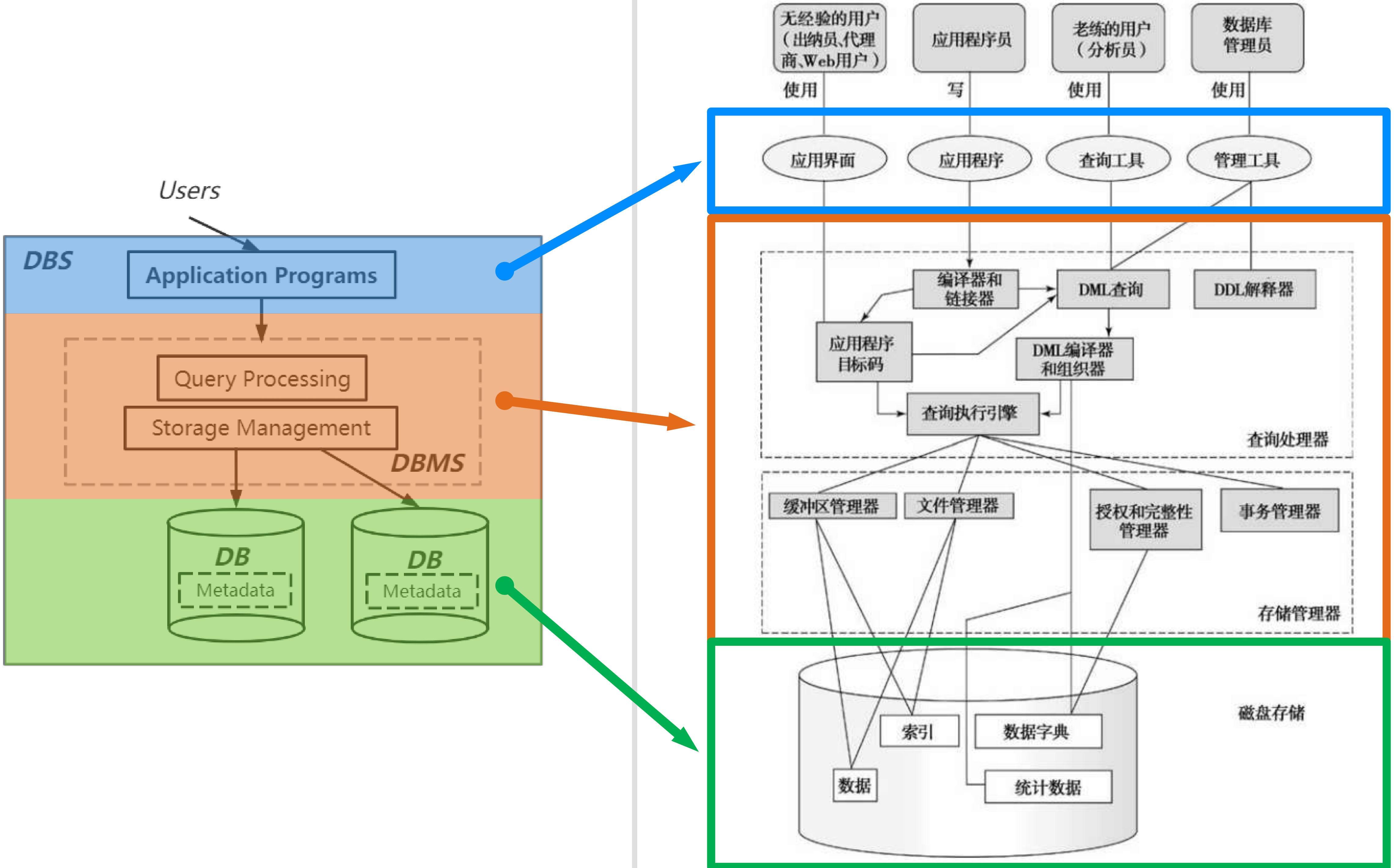
ubuntu: dpkg --get-selections | grep mysql



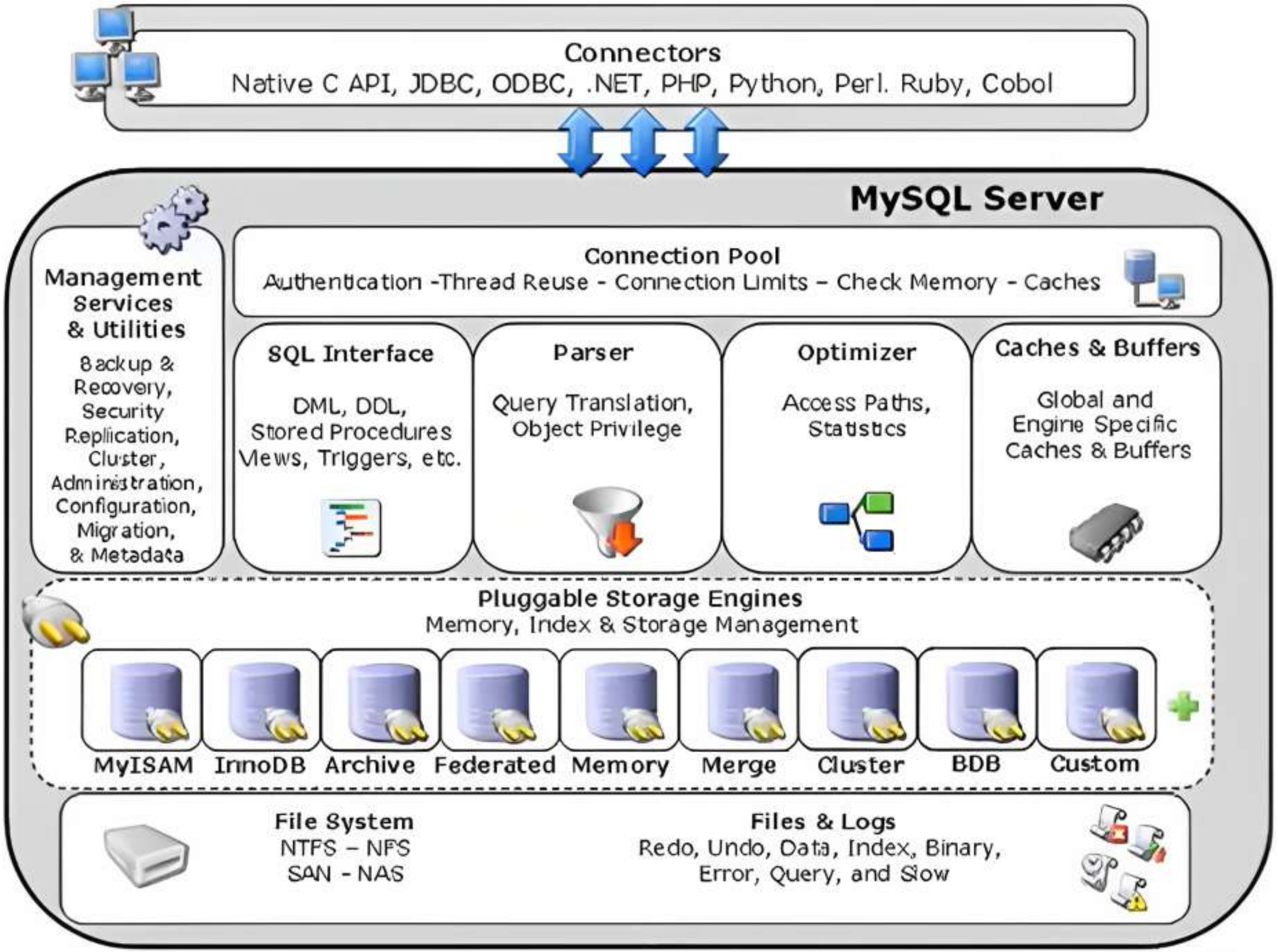
/usr/bin/mysql

/usr/sbin/mysqld

/var/lib/mysql/



Users
DBS
Application Programs
Query Processing
Storage Management
DBMS
DB
Metadata
DB
Metadata
无经验的用户（出纳员、代理商、Web用户）
应用程序员
老练的用户（分析员）
数据库管理员
使用
写
使用
使用
应用界面
应用程序
查询工具
管理工具
编译器和链接器
DML查询
DDL解释器
应用程序目标码
DML编译器和组织器
查询执行引擎
查询处理器
缓冲区管理器
文件管理器
授权和完整性管理器
事务管理器
存储管理器
磁盘存储
索引
数据字典
数据
统计数据



Connectors
Native C API, JDBC, ODBC, .NET, PHP, Python, Perl. Ruby, Cobol
MySQL Server
Management Services & Utilities
Backup & Recovery, Security Replication, Cluster, Administration, Configuration, Migration, & Metadata
Connection Pool
Authentication -Thread Reuse - Connection Limits - Check Memory - Caches
SQL Interface
DML, DDL, Stored Procedures Views, Triggers, etc.
Parser
Query Translation, Object Privilege
Optimizer
Access Paths, Statistics
Caches & Buffers
Global and Engine Specific Caches & Buffers
Pluggable Storage Engines
Memory, Index & Storage Management
MyISAM
InnoDB
Archive
Federated
Memory
Merge
Cluster
BDB
Custom
File System
NTFS – NFS
SAN - NAS
Files & Logs
Redo, Undo, Data, Index, Binary, Error, Query, and Slow

数据管理

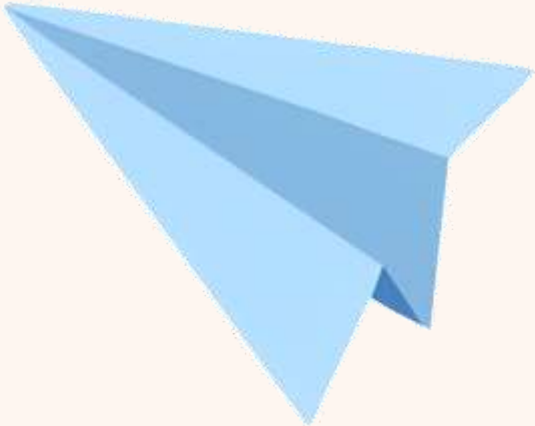
数据库系统

DBMS 的特色

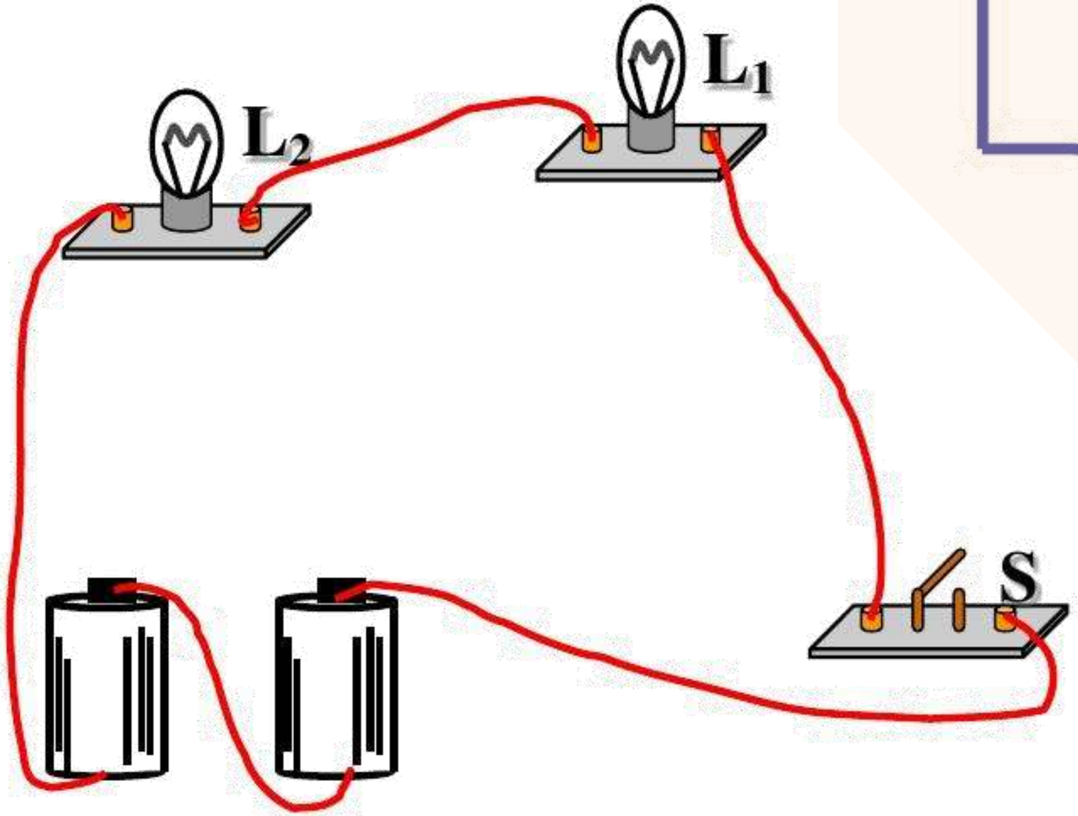
DBMS的特色

- 数据独立性
- 数据库语言
- 事务处理

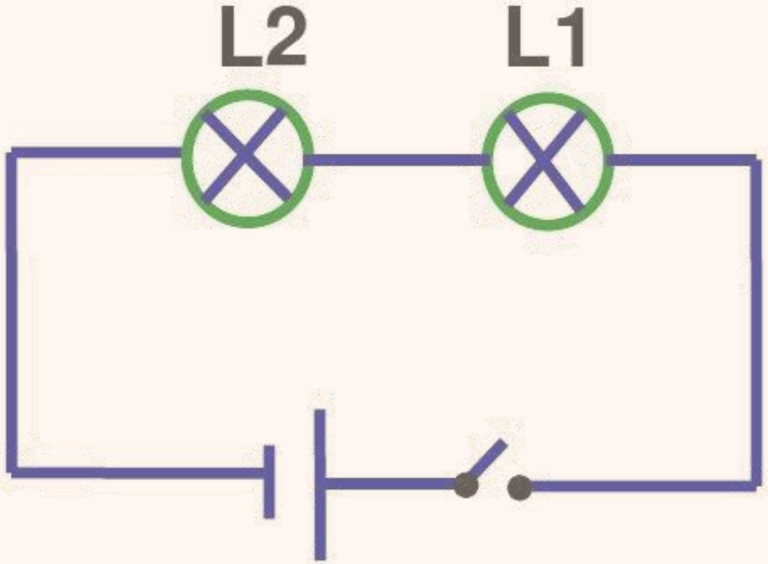
数据模型



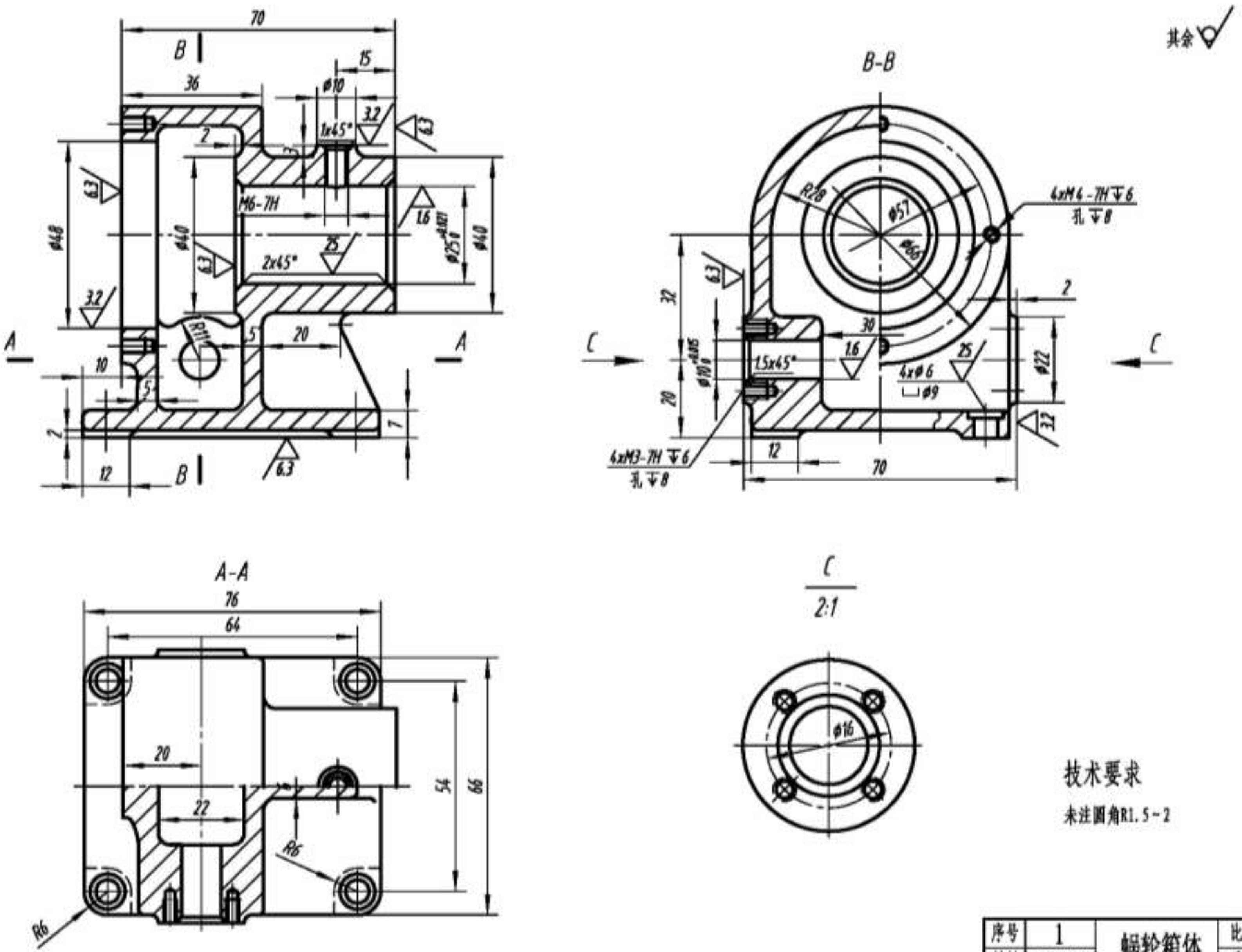

WFLY
WFT09
1024



L2
L1
S



L2
L1

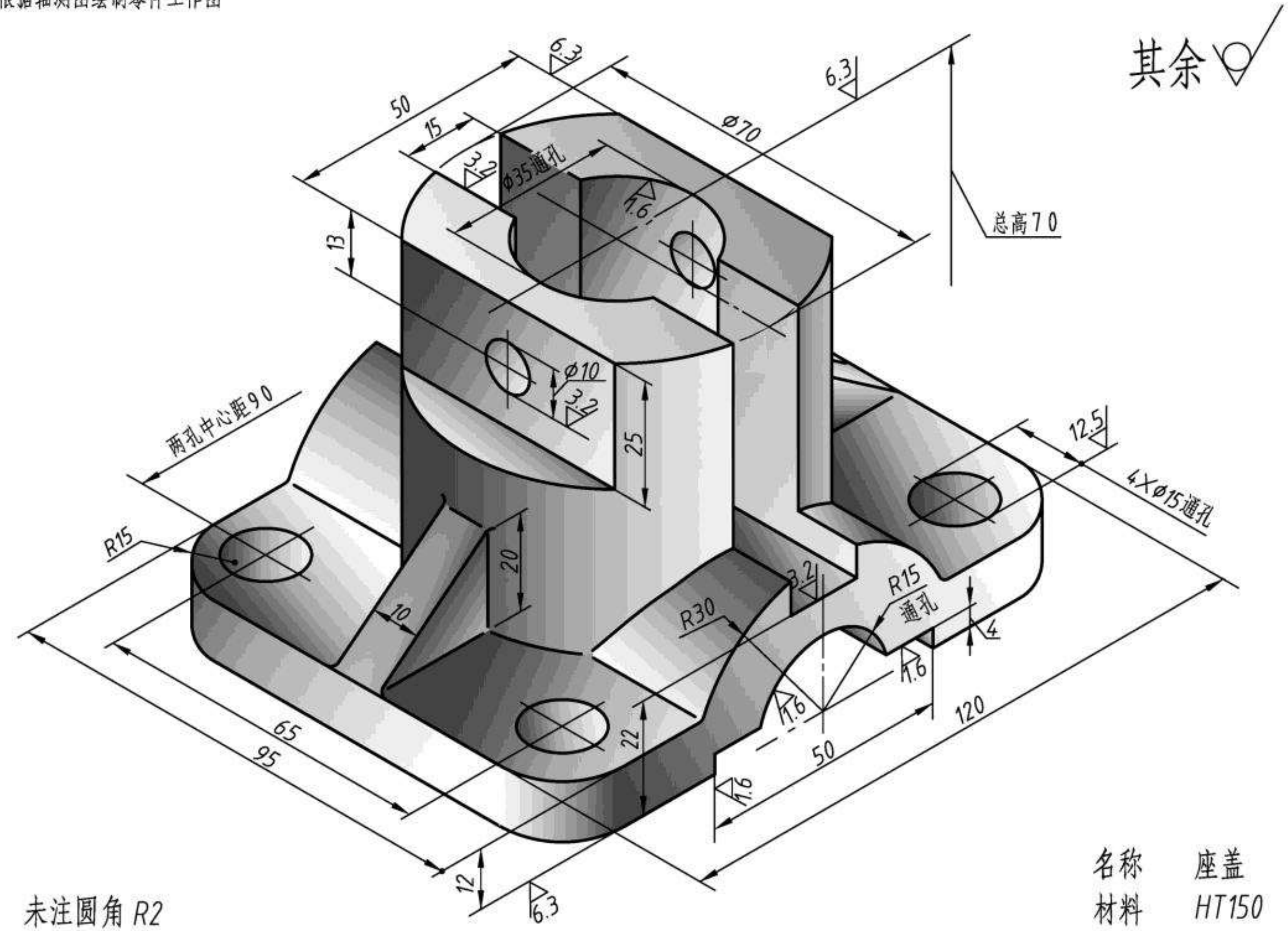


技术要求

未注圆角R1.5~2

序号	1	蜗轮箱体	比 例	1:1.5
材料	HT18-36		重 量	
制图		(单位)		

10-1 根据轴测图绘制零件工作图

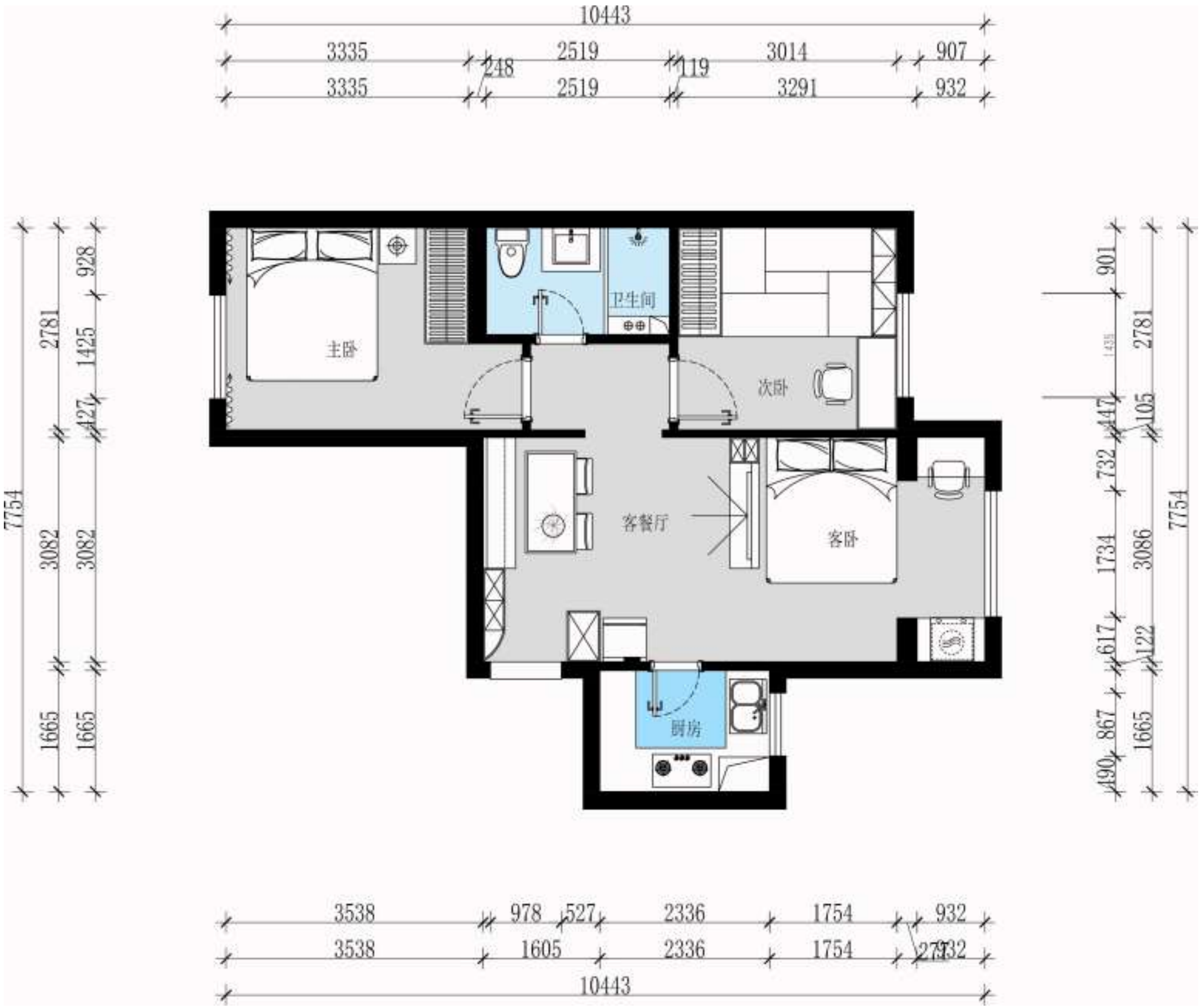


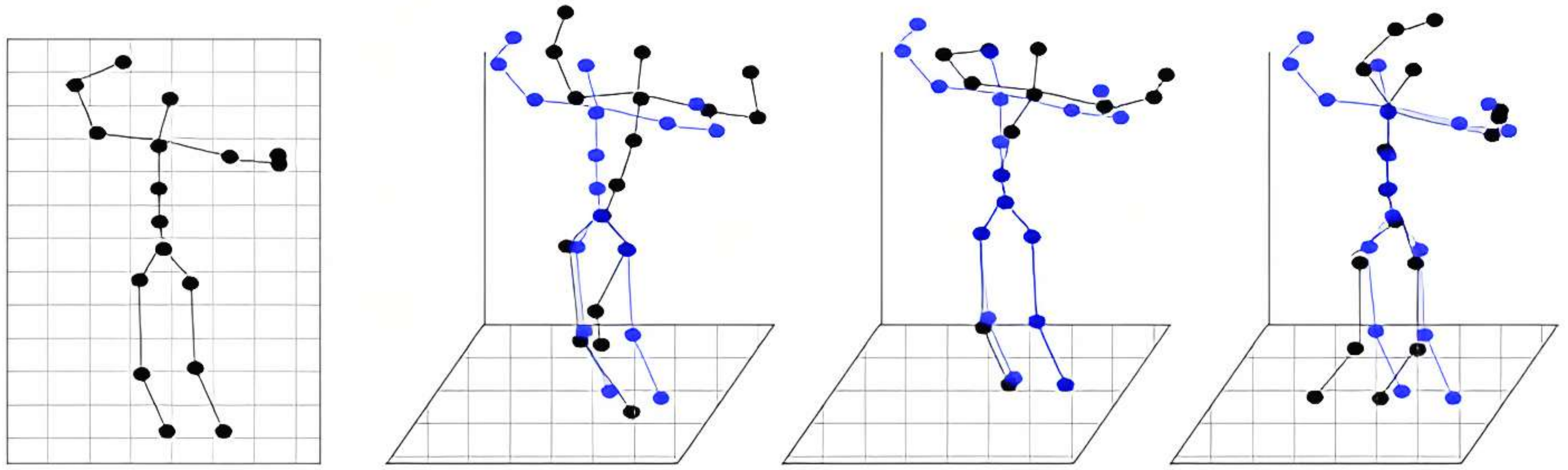
未注圆角 *R2*

名称　座盖

材料　*HT150*

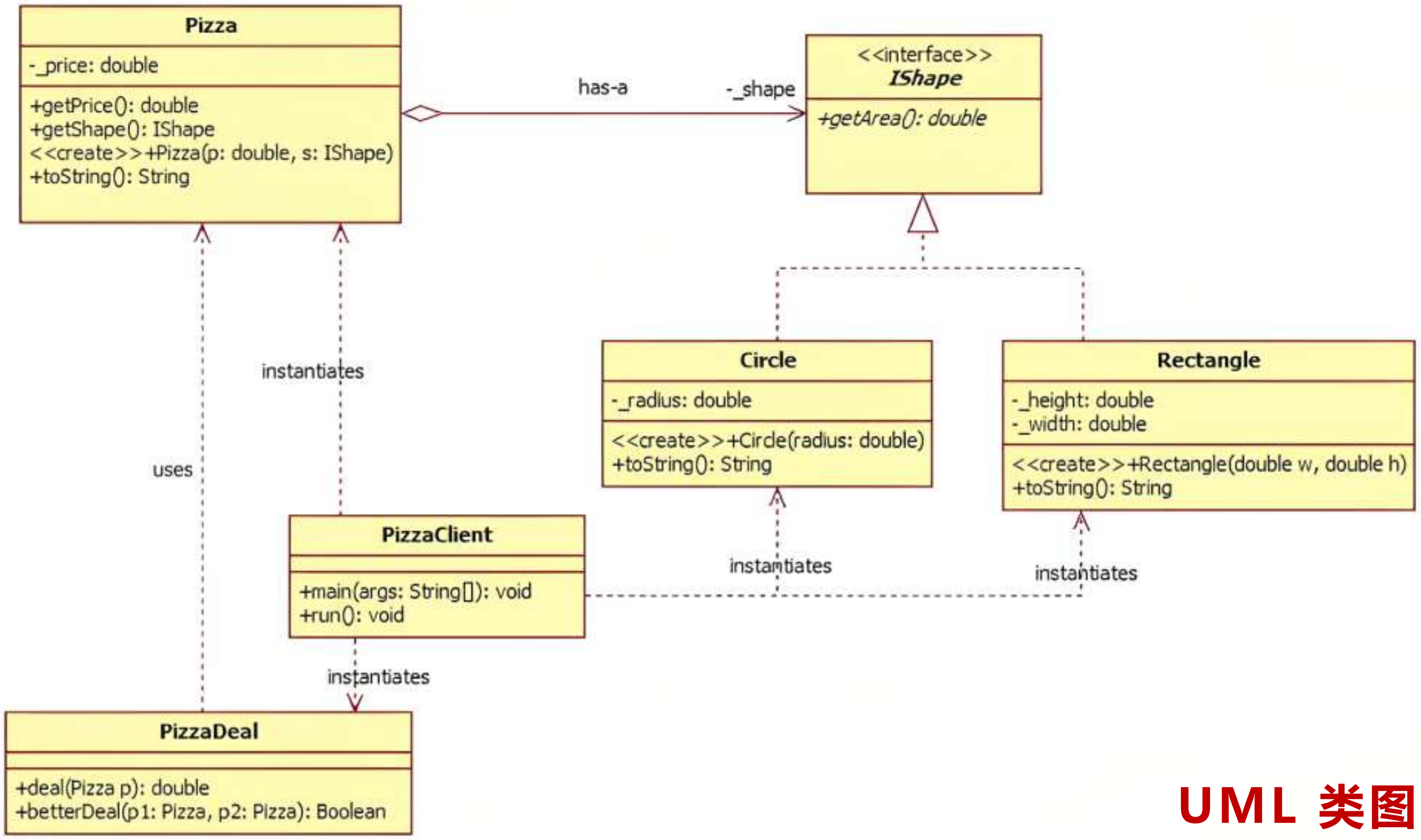
	第十章　二维的零件图和装配图表达	班级		姓名		日期	


10443
3335
2519
3014
907
248
119
3335
2519
3291
932
主卧
卫生间
次卧
客餐厅
客卧
厨房
928
1425
427
2781
3082
3082
7754
1665
1665
901
447
105
2781
732
1734
3086
617
122
867
490
1665
7754
3538
978
527
2336
1754
932
3538
1605
2336
1754
932
10443



(a)

(b)



Pizza
-_price: double
+getPrice(): double
+getShape(): IShape
<<create>>+Pizza(p: double, s: IShape)
+toString(): String
has-a
-_shape
<<interface>>
IShape
+getArea(): double
instantiates
uses
Circle
-_radius: double
<<create>>+Circle(radius: double)
+toString(): String
Rectangle
-_height: double
-_width: double
<<create>>+Rectangle(double w, double h)
+toString(): String
PizzaClient
+main(args: String[]): void
+run(): void
instantiates
instantiates
instantiates
PizzaDeal
+deal(Pizza p): double
+betterDeal(p1: Pizza, p2: Pizza): Boolean

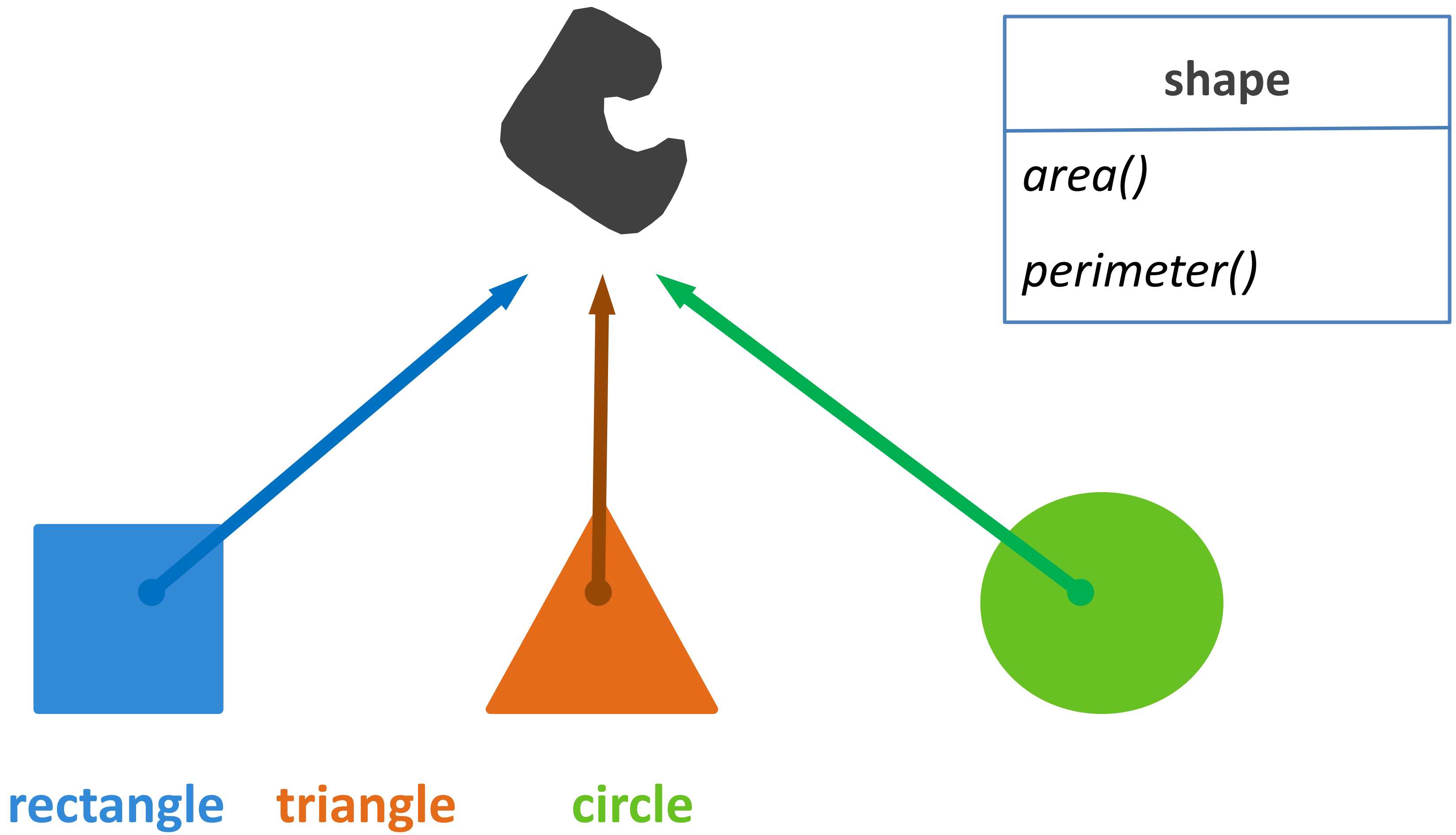
UML 类图

■ 模型（Model）

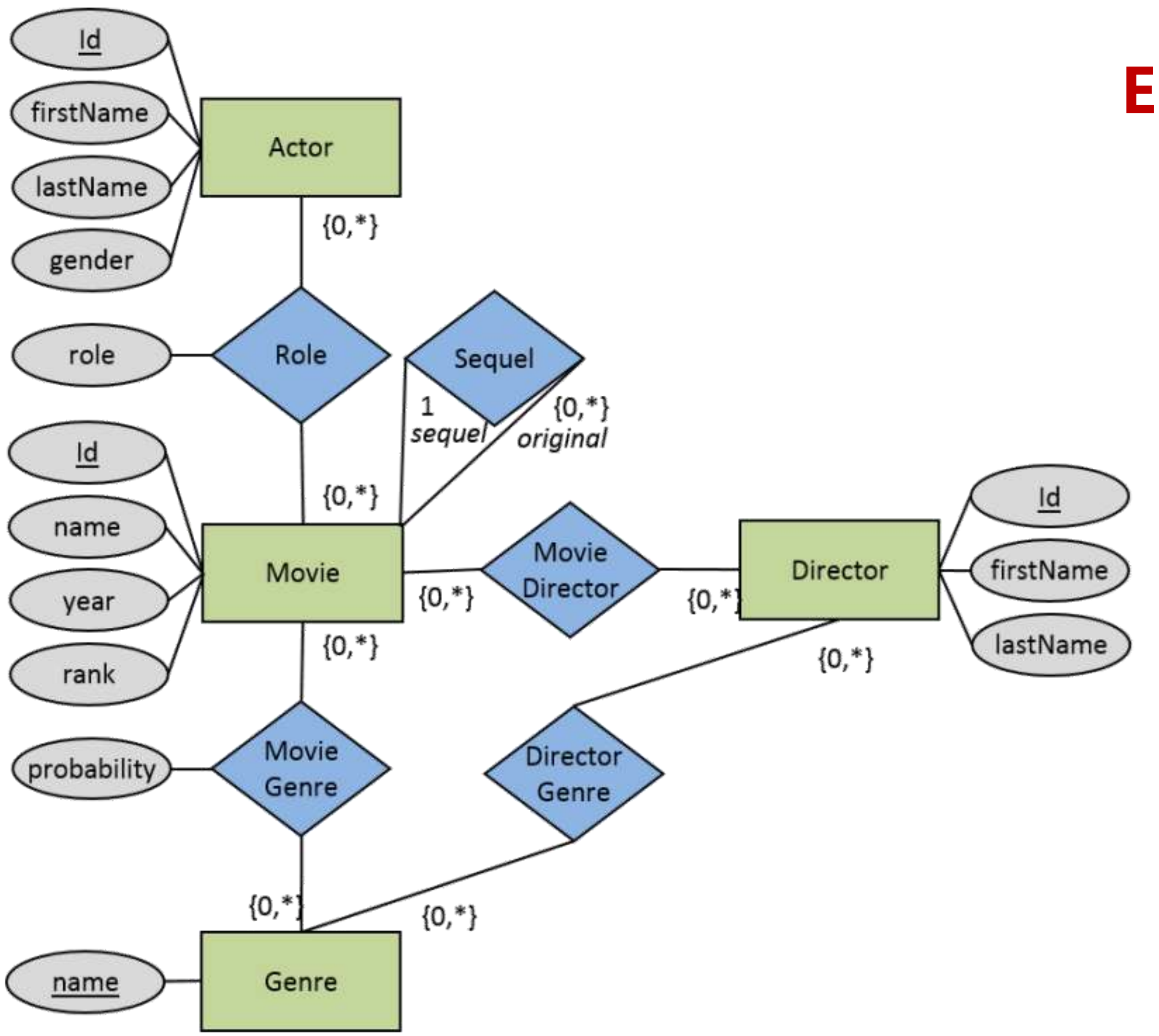
模型是对研究对象**进行抽象的工具**，抽象的结果通常是可视化的图形。

■ 建模（Modeling）

建模是对研究对象进行**抽象**的过程。抽象就是对事物进行**简化**、**提取特征**或**共性**，实现模型和事物之间的**映射或投影**，等操作。


shape
area()
perimeter()
rectangle
triangle
circle

E-R 图



Id
firstName
lastName
gender
Actor
{0,*}
role
Role
Sequel
1
sequel
{0,*}
original
Id
name
year
rank
{0,*}
Movie
Movie
Director
{0,*}
{0,*}
Director
Id
firstName
lastName
{0,*}
{0,*}
probability
Movie
Genre
Director
Genre
{0,*}
{0,*}
name
Genre

数据模型（Data Model）

数据模型是完成**数据抽象的工具**，即，用来描述**数据**、数据**联系**、数据**语义**以及一致性**约束**的一套概念工具。

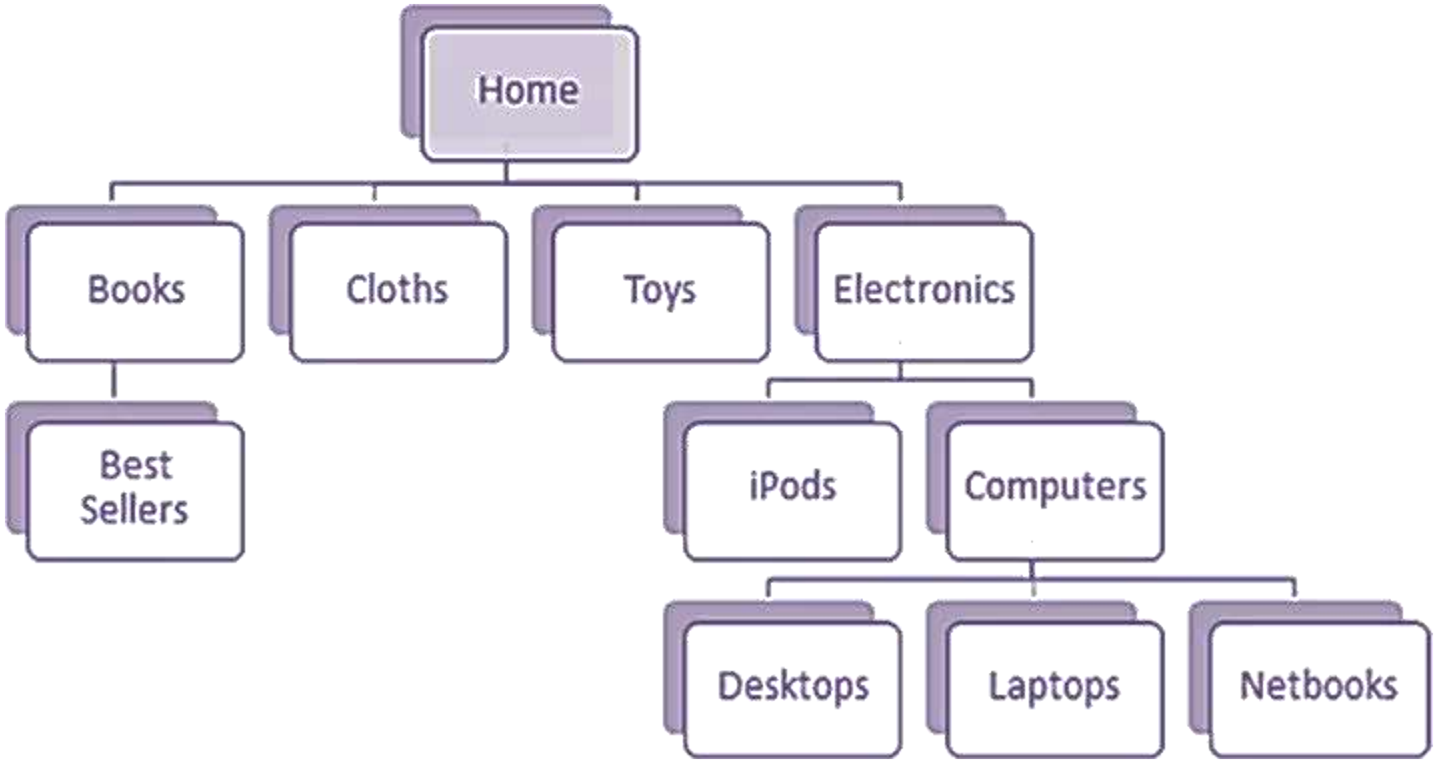
数据模型分类

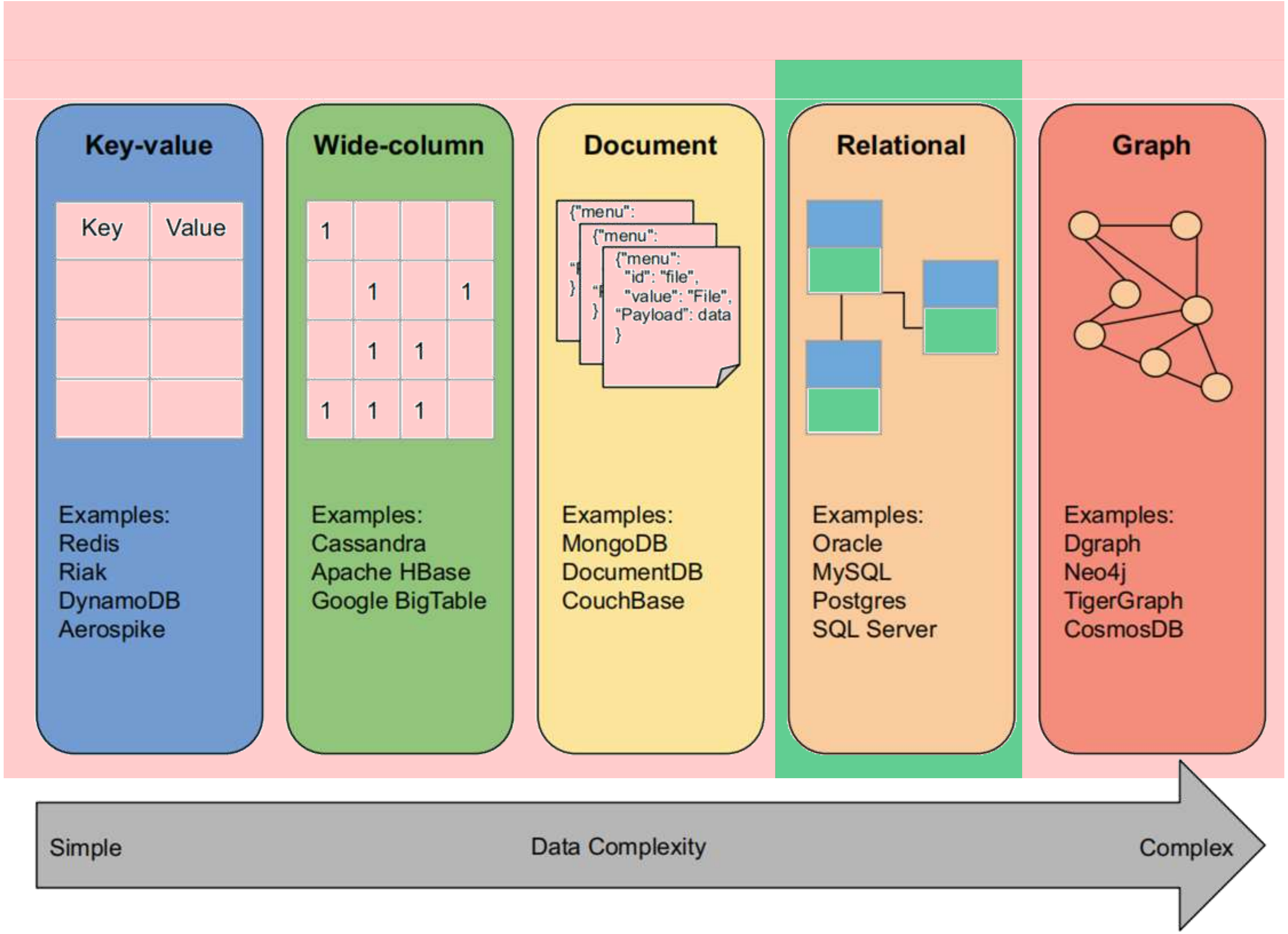
概念模型： **<u>E-R 模型</u>**，UML 类模型

逻辑模型：	层次模型	50 末	IMS
	网络模型	60 中	Honeywell
	<u>关系模型</u>	70 初	MySQL
	对象模型	80 末	db4o
	半结构化模型	00 初	BaseX
	……		

物理模型： Heap File 模型，**<u>ISAM 模型</u>**，…

层次模型	网络模型	关系模型	…

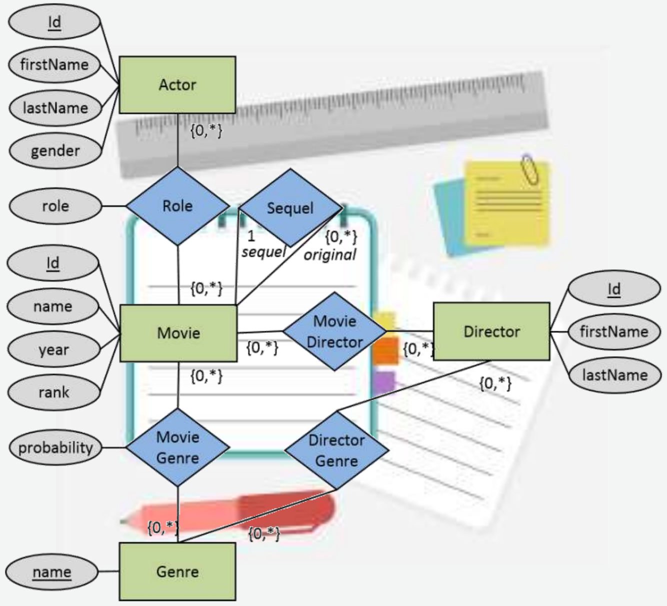




Key-value
Key
Value
Examples:
Redis
Riak
DynamoDB
Aerospike
Wide-column
1
1
1
1
1
1
1
1
Examples:
Cassandra
Apache HBase
Google BigTable
Document
{"menu":
{"menu":
{"menu":
"id": "file",
"value": "File",
"Payload": data
}
Examples:
MongoDB
DocumentDB
CouchBase
Relational
Examples:
Oracle
MySQL
Postgres
SQL Server
Graph
Examples:
Dgraph
Neo4j
TigerGraph
CosmosDB
Simple
Data Complexity
Complex

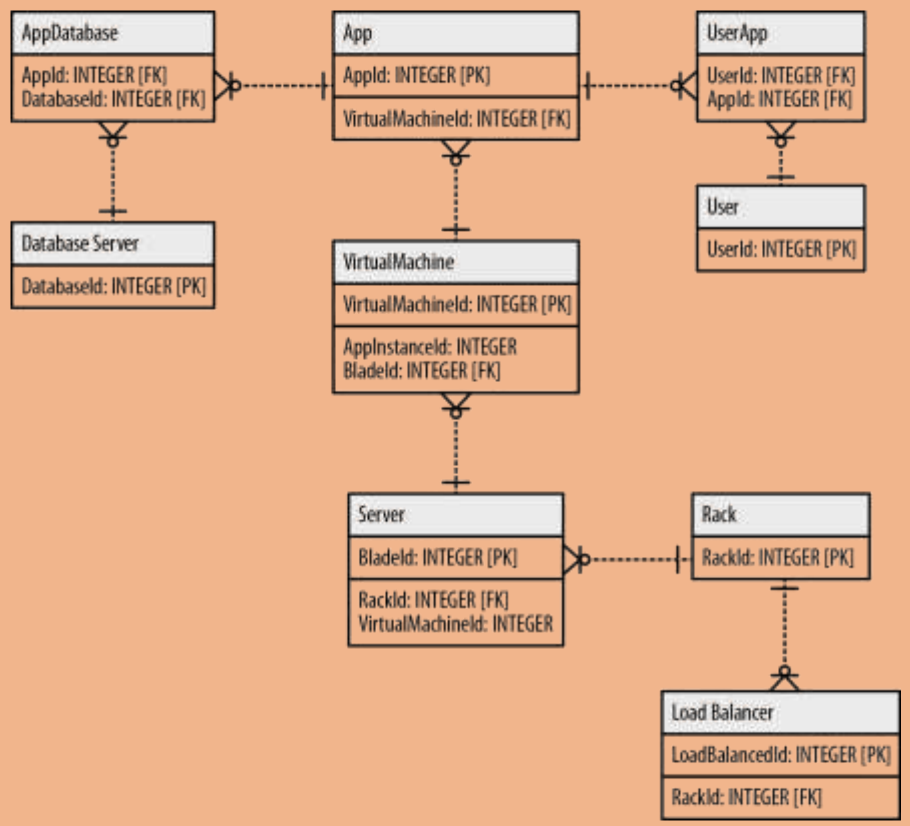
概念层

概念模型



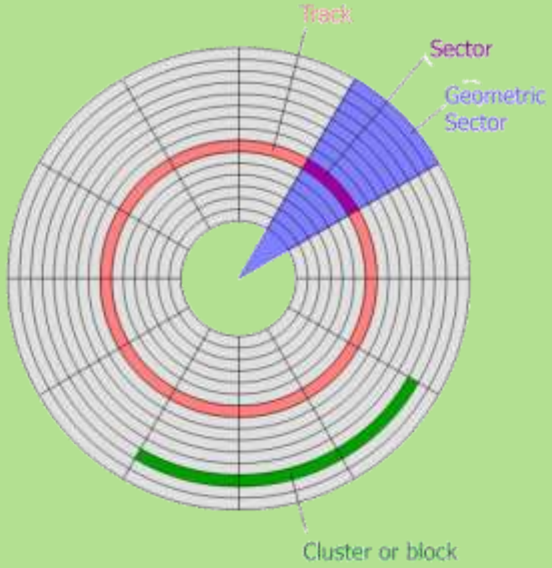
逻辑层

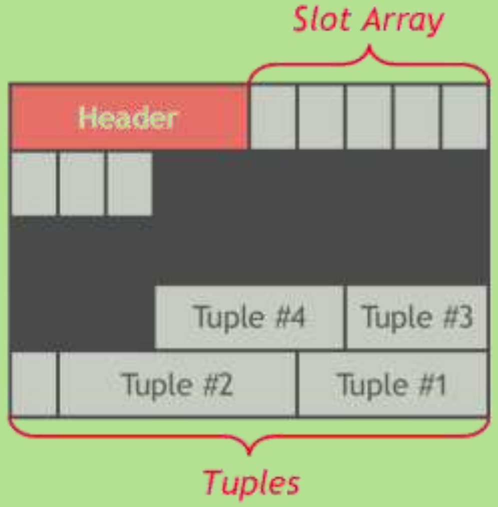
逻辑模型



物理层

物理模型





数据库模式

模式（Schema）和实例（Instance）

```
int age = 17;                                简单类型

struct instructor {                          复杂类型
  int id;
  char name[20];
  char dept_name[20];
  float salary;
};

struct instructor t1;

t1.id = 1000;
strcpy(t1.name, "wangding");
strcpy(t1.dept_name, "IT");
t1.salary = 6500;
```

类型 $\left\{ \begin{array}{l} \text{简单类型} \\ \text{复杂类型} \end{array} \right.$

类型 $\left\{ \begin{array}{l} \text{内置类型} \\ \text{自定义类型} \end{array} \right.$

类型的作用 $\left\{ \begin{array}{l} \text{存储空间大小} \\ \text{数据取值范围或精度} \\ \text{能够执行的操作} \end{array} \right.$

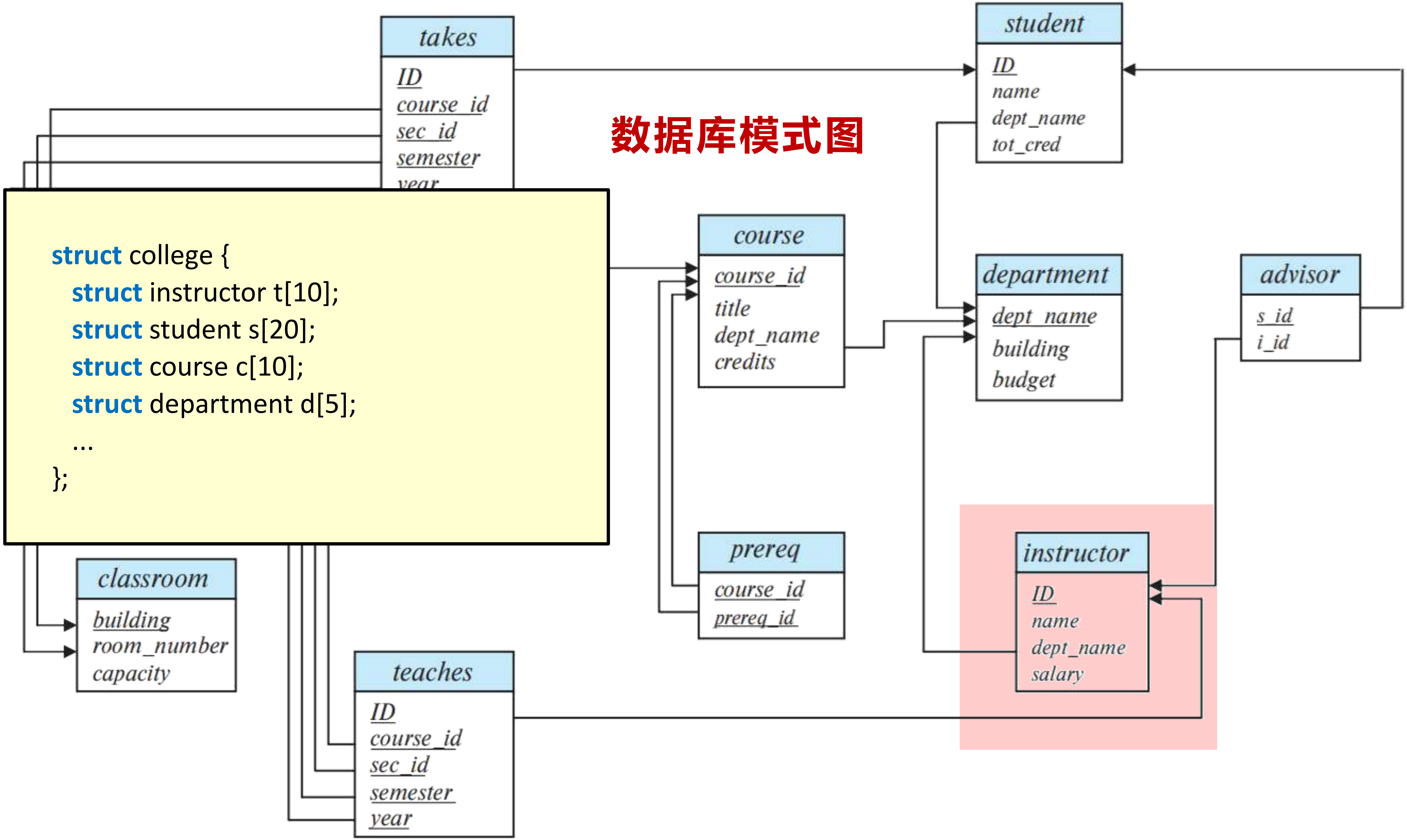
■ 模式（Schema）和实例（Instance）

```
create table instructor
    (ID          varchar (5),
     name        varchar (20) not null,
     dept_name   varchar (20),
     salary      numeric (8,2),
     primary key (ID),
     foreign key (dept_name) references department);
```

```
struct instructor {
   int id;
   char name[20];
   char dept_name[20];
   float salary;
};

struct instructor t[12];
```

ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

数据库模式图

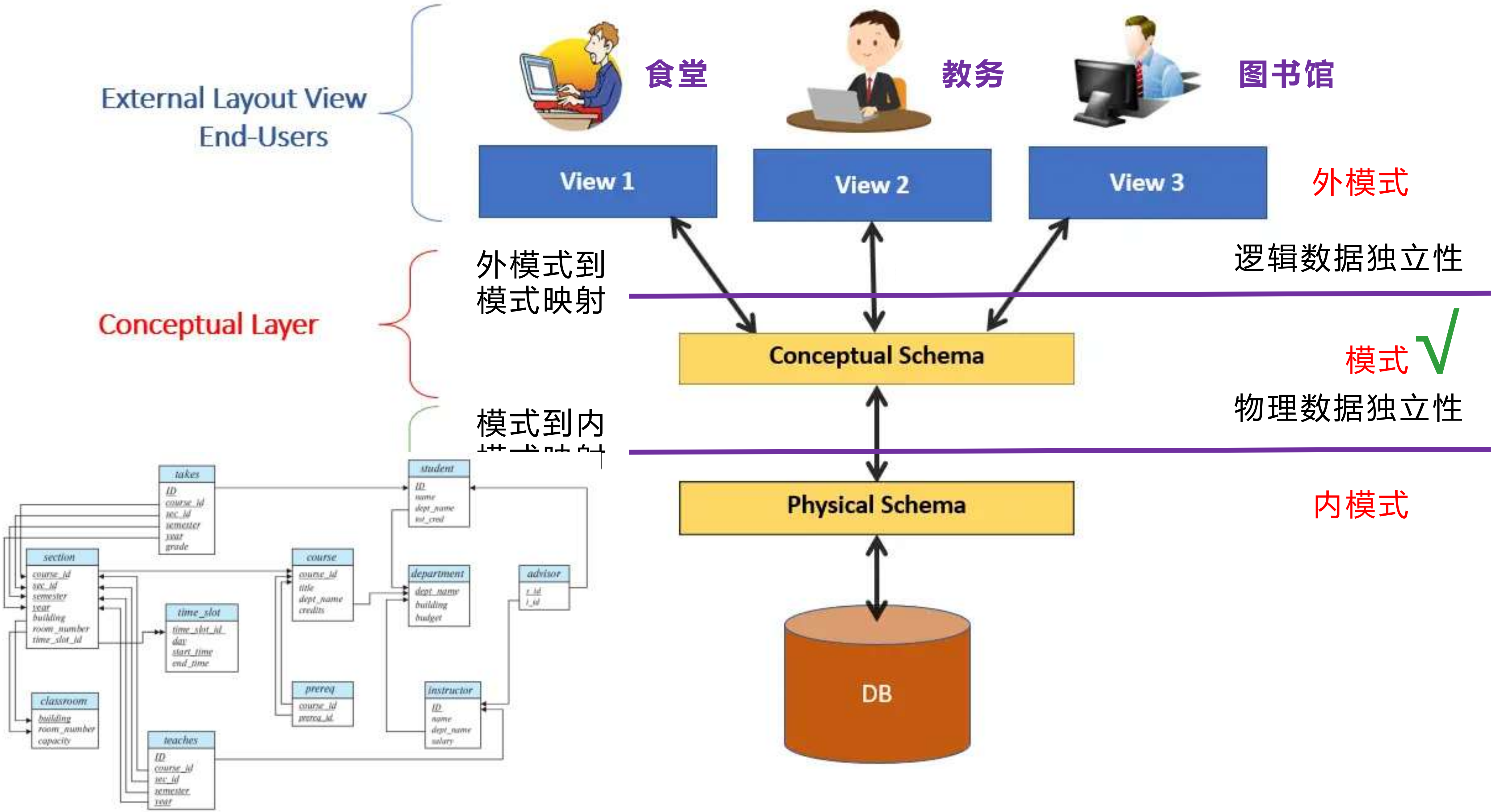



```
struct college {
  struct instructor t[10];
  struct student s[20];
  struct course c[10];
  struct department d[5];
  ...
};
```

模式（Schema）和实例（Instance）

模式是数据的**定义和描述**，实例则是模式在某一时刻的具体数据。

模式是元数据
数据是变化的，模式是不变的
数据库设计，数据库建模，实体 → 模式
数据库使用，创建数据库，模式 → 元组
模式的层次，属性模式？关系模式，数据库模式
模式 = 类型 + 约束 + 联系



External Layout View
End-Users
食堂
教务
图书馆
View 1
View 2
View 3
外模式
外模式到
模式映射
逻辑数据独立性
Conceptual Layer
Conceptual Schema
模式 √
模式到内
物理数据独立性
Physical Schema
内模式
DB
takes
student
section
course
department
advisor
time_slot
classroom
prereq
instructor
teaches

■ 数据独立性

- 抽象 → 模型
  - 操作
    - 去掉个性
    - 保留共性
    - 进行投影
  - 作用
    - 化简（屏蔽复杂性）
    - 抗变化
  - 结果
- **<u>数据模型</u>**
  - 概念模型
  - 逻辑模型
    - 层次模型
    - 网状模型
    - 关系模型
    - …
  - 物理模型
- 类型
- **<u>数据库模式</u>**
  - 外模式、用户模式、子模式
  - 模式、概念模式、数据库模式
  - 内模式、存储模式、物理模式

Person

Login	LastName	FirstName
skol	Kovalevskaya	Sofia
mlom	Lomonosov	Mikhail
dmitri	Mendeleev	Dmitri
ivan	Pavlov	Ivan

Project

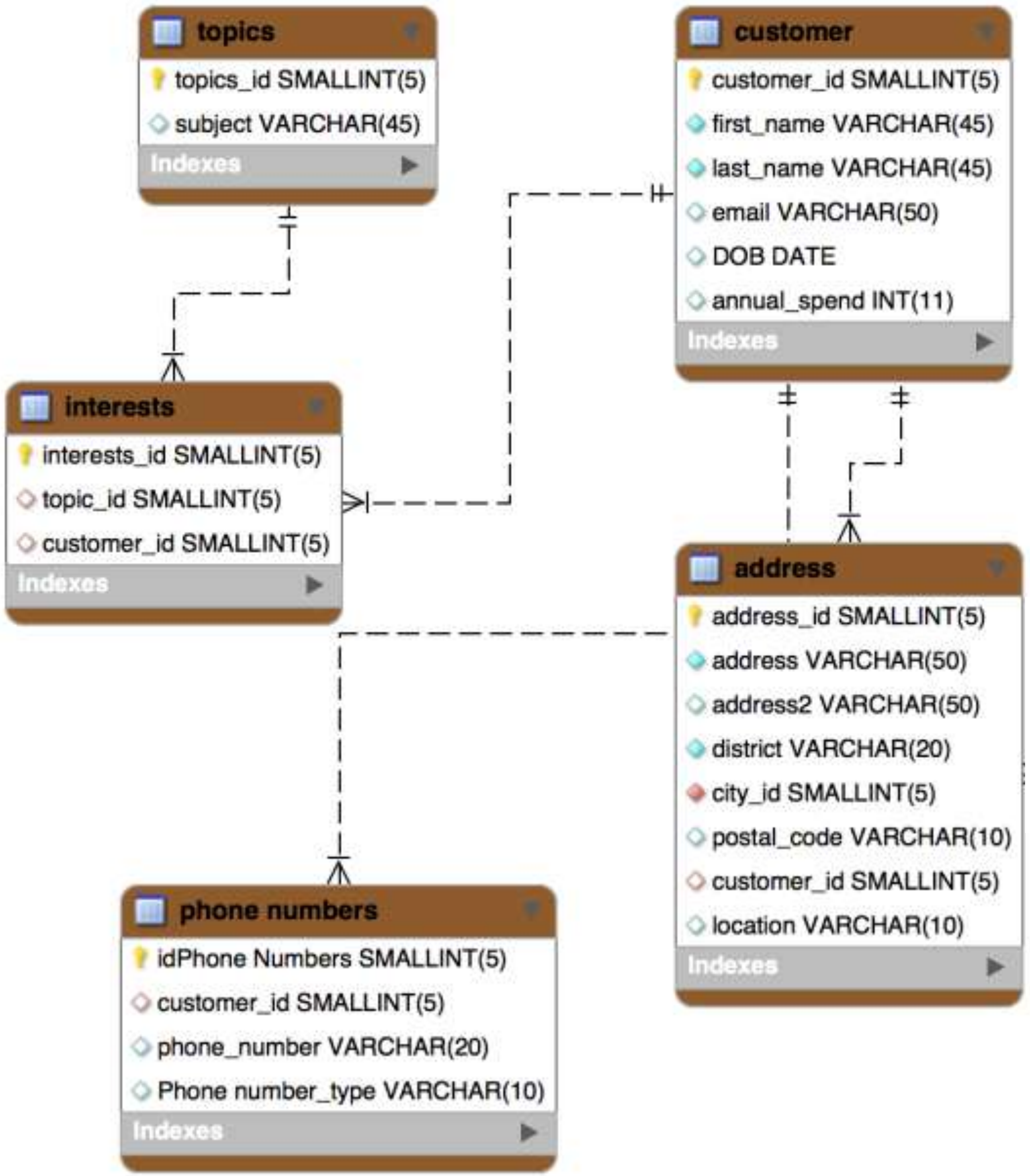
ProjectId	ProjectName
1214	Antigravity
1709	Teleportation
1737	Time Travel

Experiment

ProjectId	ExperimentId	NumInvolved	ExperimentDate	Hours
1214	1	1	NULL	1.5
1214	2	1	1889-11-01	14.3
1709	1	3	1891-01-22	7.0
1709	2	1	1891-02-23	7.2
1737	1	1	1900-07-05	-1.0
1737	2	2	1900-07-05	-1.5

Involved

ProjectId	ExperimentId	InvolvedId	Login
1214	1	1	mlom
1214	2	1	mlom
1709	1	1	dmitri
1709	1	2	skol
1709	1	3	ivan
1709	2	1	mlom
1737	1	1	skol
1737	2	1	skol
1737	2	2	ivan



■ DBMS的特色

- ➢ 数据独立性
- ➢ 数据库语言
- ➢ 事务处理

编程语言：SQL	第 2 章
数学语言：关系代数	第 3 章
设计语言：E-R 图	第 4 章

SQL: Structured Query Language

E-R: Entity-Relationship

7w

- 第1章 绪论
- 第2章 结构化查询语言
- 第3章 关系代数

3w

- 第4章 概念数据库设计
- 第5章 逻辑数据库设计
- 第6章 物理数据库设计

6w

- 第7章 存储管理
- 第8章 索引结构
- 第9章 查询执行
- 第10章 查询优化
- 第11章 并发控制
- 第12章 故障恢复

编程语言：SQL

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

```sql
select books.id, books.unique_id, authors.author_name
from books, authors, author_books ab
where books.id = ab.book_id
and authors.id = ab.author_id;
```

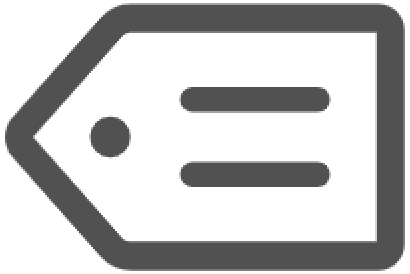
SQL 代码

id	unique_id	author_name
1	yao-shen-ji	踏雪动漫
2	yuan-zun	编绘：Dr.大吉
3	yuan-zun	原著：天蚕土豆
4	ni-tian-jian-shen	Amber漫研社
5	wu-dao-du-zun	破界互动
6	wu-shen-zhu-zai	踏雪动漫
7	ling-jian-zun	iCiyuan动漫 留下
8	shen-wu-tian-zun	绍兴凰漫
9	yu-ling-shi	时代漫王
10	fei-ren-zai	一汪空气
11	tao-hua-bao-dian	有鹿文化
12	jiu-yang-shen-wang	掌阅漫画
13	dou-luo-da-lu	风炫动漫
14	xing-wu-shen-jue	踏雪动漫

Read Only

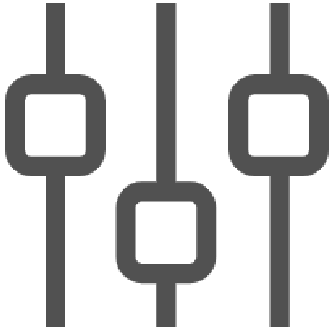
Output

Action Output



DDL

DML



DCL

SQL: Structured Query Language

增：create

删：drop

改：alter

DDL: Data Definition Language

```sql
drop database if exists test;
create database test default character set utf8mb4;

drop table if exists areas;
create table areas (
  id int(11) not null auto_increment,
  area_name varchar(255) not null,
  primary key (id),
  unique key area_name (area_name)
) engine=innodb default charset=utf8mb4;

```

DDL

增：insert

删：delete

改：update

查：select

DML: Data Manipulation Language

MySQL Workbench

```sql
select books.id, books.unique_id, authors.author_name
from books, authors, author_books ab
where books.id = ab.book_id
and authors.id = ab.author_id;
```

DML

id	unique_id	author_name
1	yao-shen-ji	踏雪动漫
2	yuan-zun	编绘：Dr.大吉
3	yuan-zun	原著：天蚕土豆
4	ni-tian-jian-shen	Amber漫研社
5	wu-dao-du-zun	破界互动
6	wu-shen-zhu-zai	踏雪动漫
7	ling-jian-zun	iCiyuan动漫 留下
8	shen-wu-tian-zun	绍兴凰漫
9	yu-ling-shi	时代漫王
10	fei-ren-zai	一汪空气
11	tao-hua-bao-dian	有鹿文化
12	jiu-yang-shen-wang	掌阅漫画
13	dou-luo-da-lu	风炫动漫
14	xing-wu-shen-jue	踏雪动漫

book_author 1

增：grant

删：revoke

DCL: Data Control Language

SQL 分类

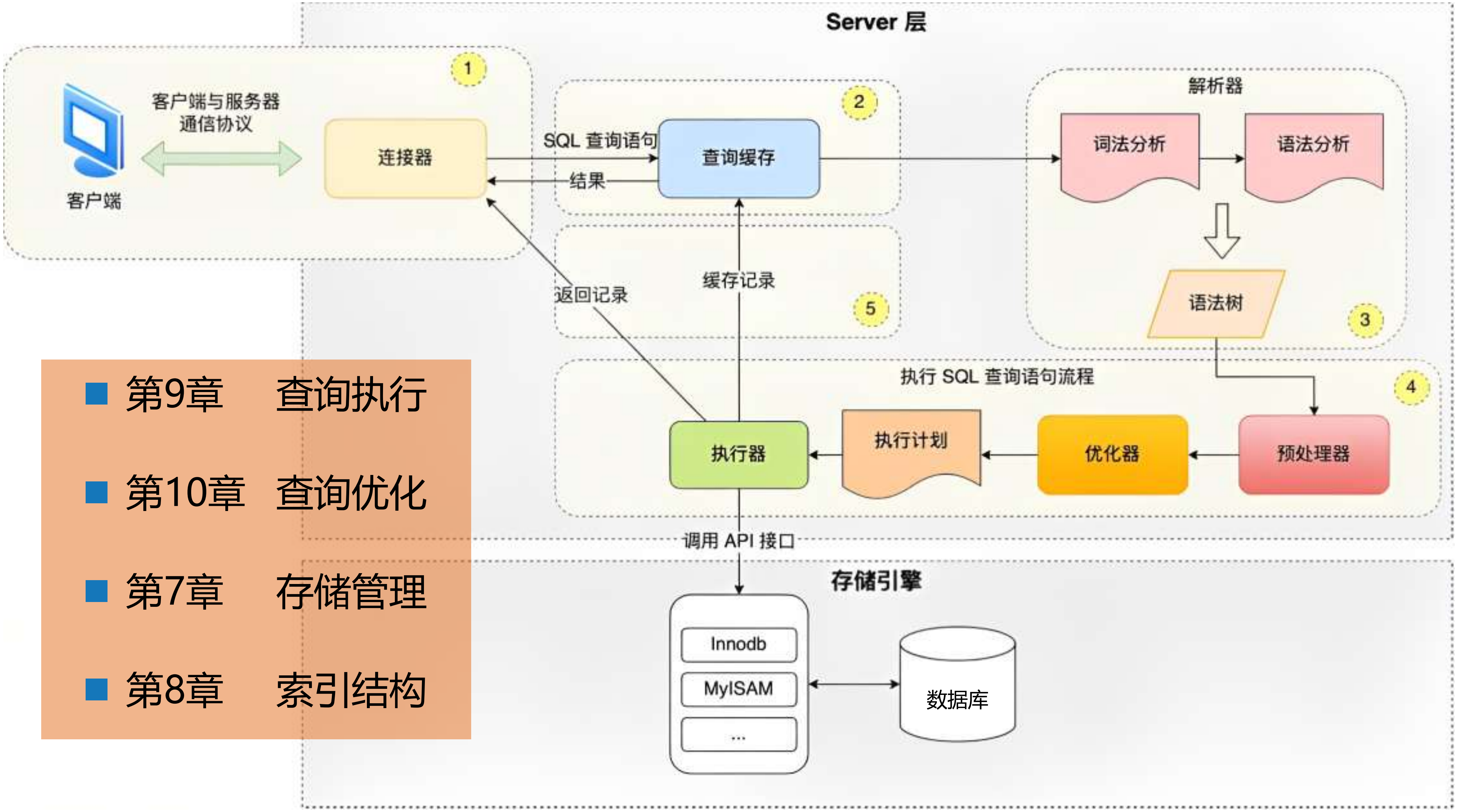
- **DDL**： Data Definition Language
- **DML**： Data Manipulation Language
- **DCL**： Data Control Language
- **DQL**： Data Query Language

- 第2章 结构化查询语言

- **DTL**： Data Transaction Language

- 第11章 并发控制
- 第12章 故障恢复

- 第9章　查询执行
- 第10章　查询优化
- 第7章　存储管理
- 第8章　索引结构



数学语言：关系代数

查询计算机系学生的选课情况，列出学号、姓名、课号、得分

$$\Pi_{Student.Sno,Sname,Cno,Grade}(\sigma_{Sdept='CS'}(Student \bowtie_{Student.Sno=SC.Sno} SC))$$

关系代数

$Student \bowtie_{Student.Sno=SC.Sno} SC$

Student.Sno	Sname	Ssex	Sage	Sdept	SC.Sno	Cno	Grade
PH-001	Nick	M	20	Physics	PH-001	1002	92
PH-001	Nick	M	20	Physics	PH-001	2003	85
PH-001	Nick	M	20	Physics	PH-001	3006	88
CS-001	Elsa	F	19	CS	CS-001	1002	95
CS-001	Elsa	F	19	CS	CS-001	3006	90
CS-002	Ed	M	19	CS	CS-002	3006	80
MA-001	Abby	F	18	Math	MA-001	1002	

传统集合运算

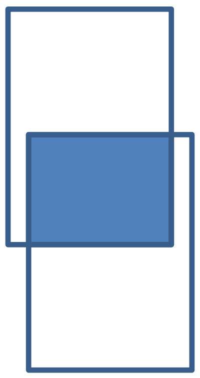
- 交：∩
- 并：∪
- 差：－
- 笛卡尔积：×

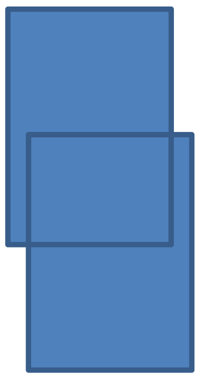
专门关系运算

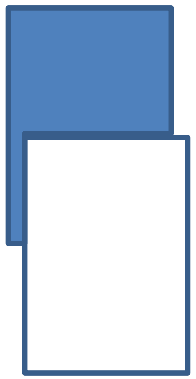
- 选择：σ
- 投影：∏
- 连接：⋈
- 除：÷

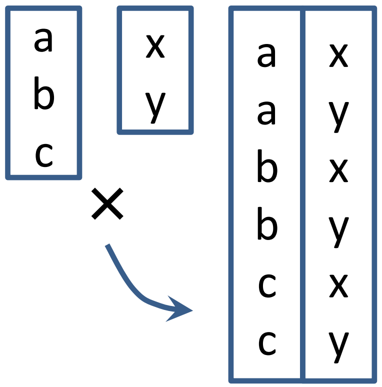
■ 传统集合运算

- 交：∩
- 并：∪
- 差：－
- 笛卡尔积：×



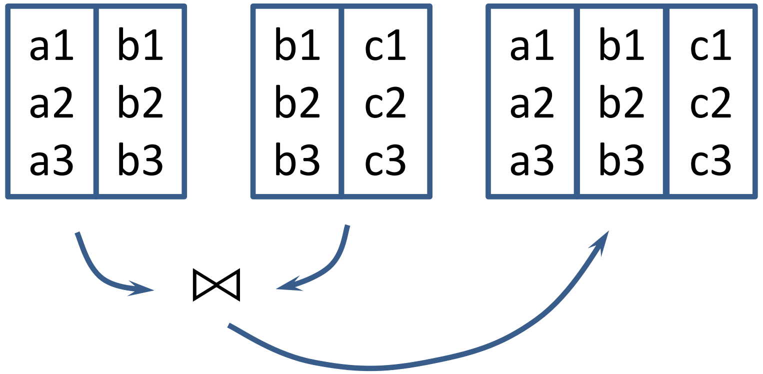


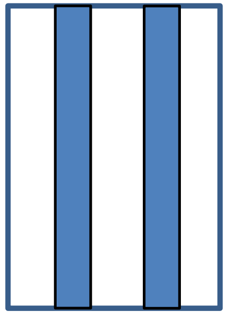


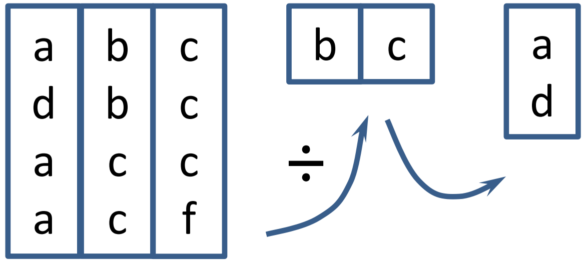


■ 专门关系运算

- 选择：σ
- 投影：∏
- 连接：⋈
- 除：÷







专门关系运算

- 选择：σ
- 投影：∏
- 连接：⋈
- 除：÷

a	α	h	η	o	ο	v	ϖ	A	Α	H	Η	O	Ο	V	ς
b	β	i	ι	p	π	w	ω	B	Β	I	Ι	P	Π	W	Ω
c	χ	j	φ	q	θ	x	ξ	C	Χ	J	ϑ	Q	Θ	X	Ξ
d	δ	k	κ	r	ρ	y	ψ	D	Δ	K	Κ	R	Ρ	Y	Ψ
e	ε	l	λ	s	σ	z	ζ	E	Ε	L	Λ	S	Σ	Z	Ζ
f	ϕ	m	μ	t	τ			F	Φ	M	Μ	T	Τ		
g	γ	n	ν	u	υ			G	Γ	N	Ν	U	Υ		

查询计算机系学生的选课情况，列出学号、姓名、课号、得分

$$\Pi_{Student.Sno,Sname,Cno,Grade}(\sigma_{Sdept='CS'}(Student \bowtie_{Student.Sno=SC.Sno} SC))$$

$$R_1 = Stuent \bowtie_{\vartheta} SC$$

$$R_2 = \sigma(R_1)$$

$$R_3 = \prod(R_2)$$

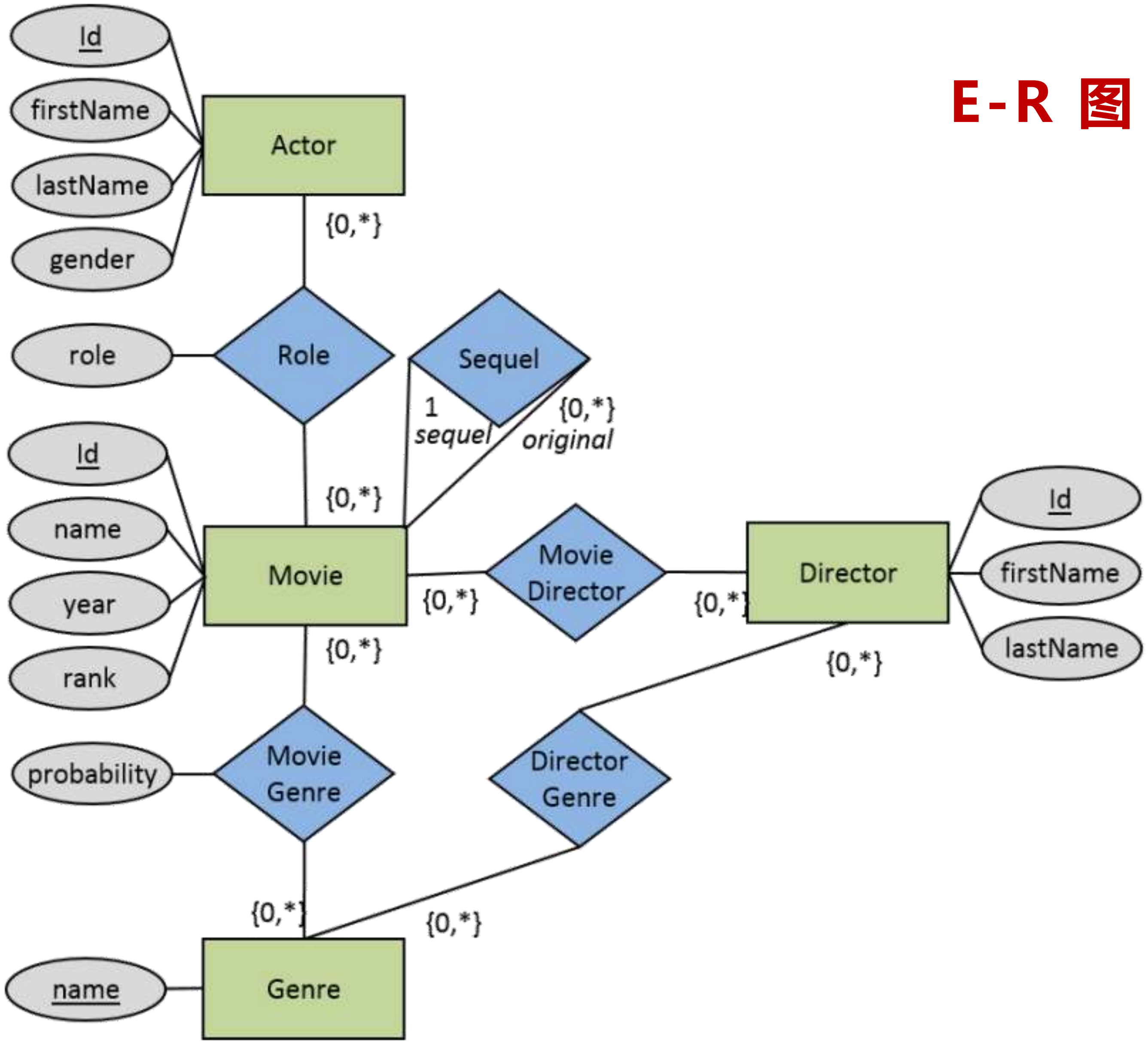
$$R_3 = \prod(\sigma(Stuent \bowtie_{\vartheta} SC))$$

- **基本关系代数运算**
  - 选择：σ
  - 投影：∏
  - 并：∪
  - 差：－
  - 笛卡尔积：×
  - 重命名：ρ

- **派生关系代数运算**
  - 交：∩
  - 自然连接：⋈
  - ϑ 连接：$\bowtie_{\vartheta}$
  - 外连接：左外⟕、右外⟖、全外⟗
  - 除：÷

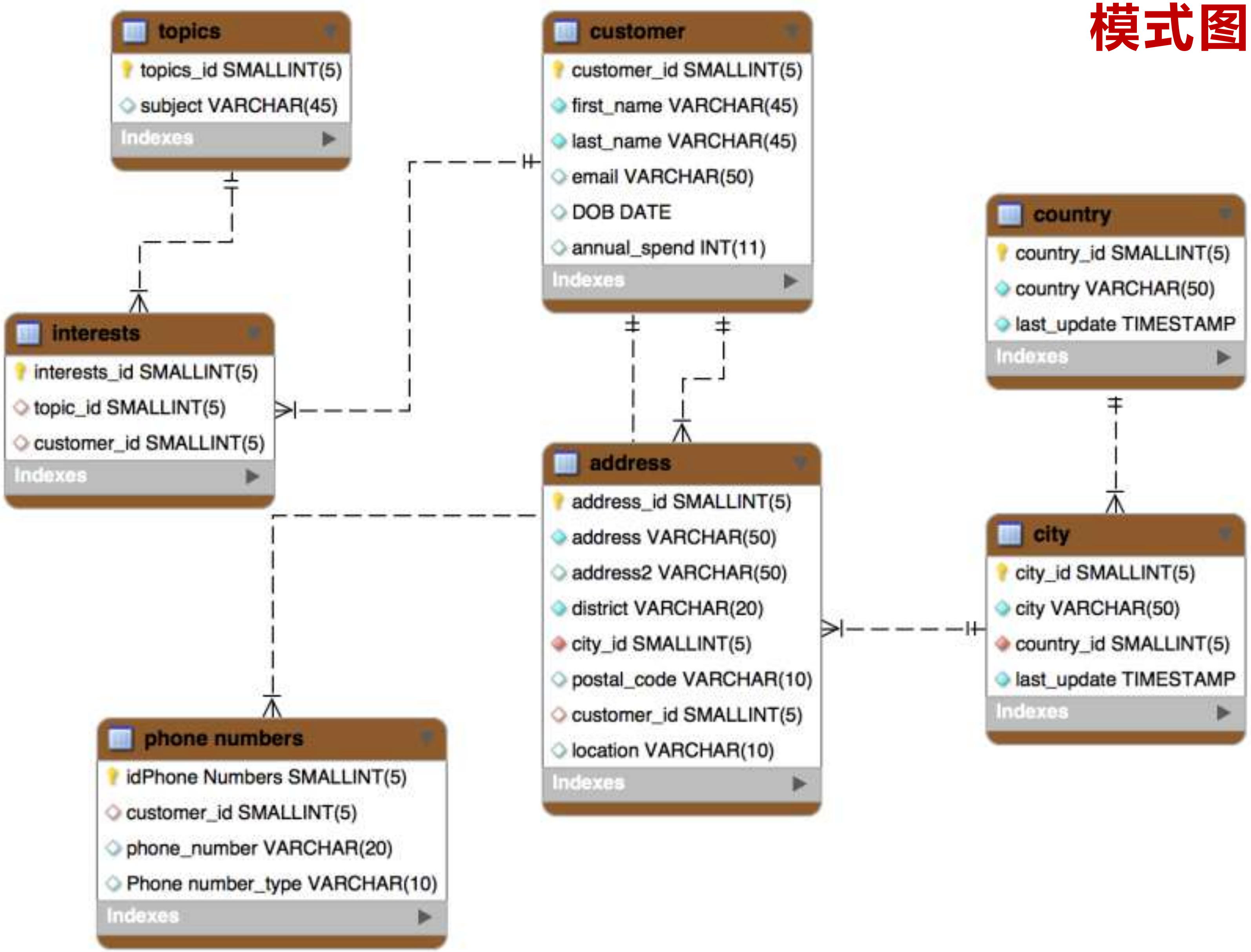
设计语言：E-R 图

E-R 图



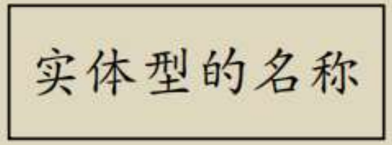
Id
firstName
lastName
gender
Actor
{0,*}
role
Role
Sequel
1
sequel
{0,*}
original
Id
name
year
rank
{0,*}
Movie
Movie
Director
{0,*}
{0,*}
Director
Id
firstName
lastName
{0,*}
{0,*}
probability
Movie
Genre
Director
Genre
{0,*}
{0,*}
name
Genre

模式图

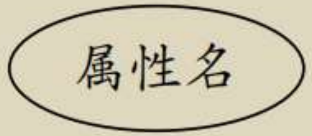

topics
topics_id SMALLINT(5)
subject VARCHAR(45)
Indexes
customer
customer_id SMALLINT(5)
first_name VARCHAR(45)
last_name VARCHAR(45)
email VARCHAR(50)
DOB DATE
annual_spend INT(11)
Indexes
country
country_id SMALLINT(5)
country VARCHAR(50)
last_update TIMESTAMP
Indexes
interests
interests_id SMALLINT(5)
topic_id SMALLINT(5)
customer_id SMALLINT(5)
Indexes
address
address_id SMALLINT(5)
address VARCHAR(50)
address2 VARCHAR(50)
district VARCHAR(20)
city_id SMALLINT(5)
postal_code VARCHAR(10)
customer_id SMALLINT(5)
location VARCHAR(10)
Indexes
city
city_id SMALLINT(5)
city VARCHAR(50)
country_id SMALLINT(5)
last_update TIMESTAMP
Indexes
phone numbers
idPhone Numbers SMALLINT(5)
customer_id SMALLINT(5)
phone_number VARCHAR(20)
Phone number_type VARCHAR(10)
Indexes

实体型的E-R图形符号

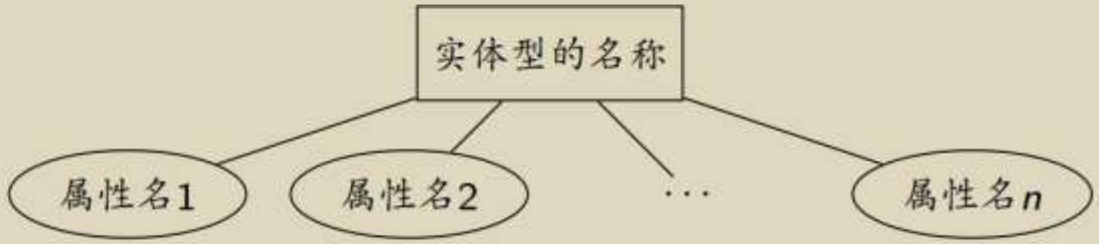
- 实体型：



- 属性：



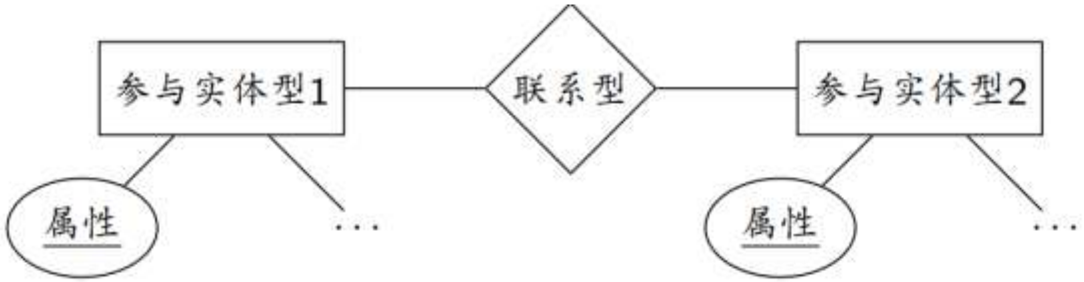
- 实体型与属性：



联系型的E-R图形符号

- 联系型：

- 实体型与联系型：



DBMS的特色

- 数据独立性
- 数据库语言
- 事务处理

■ 事务 Transaction

```
UPDATE savings_accounts
  SET balance = balance - 500
  WHERE account = 3209;
```

Decrement Savings Account

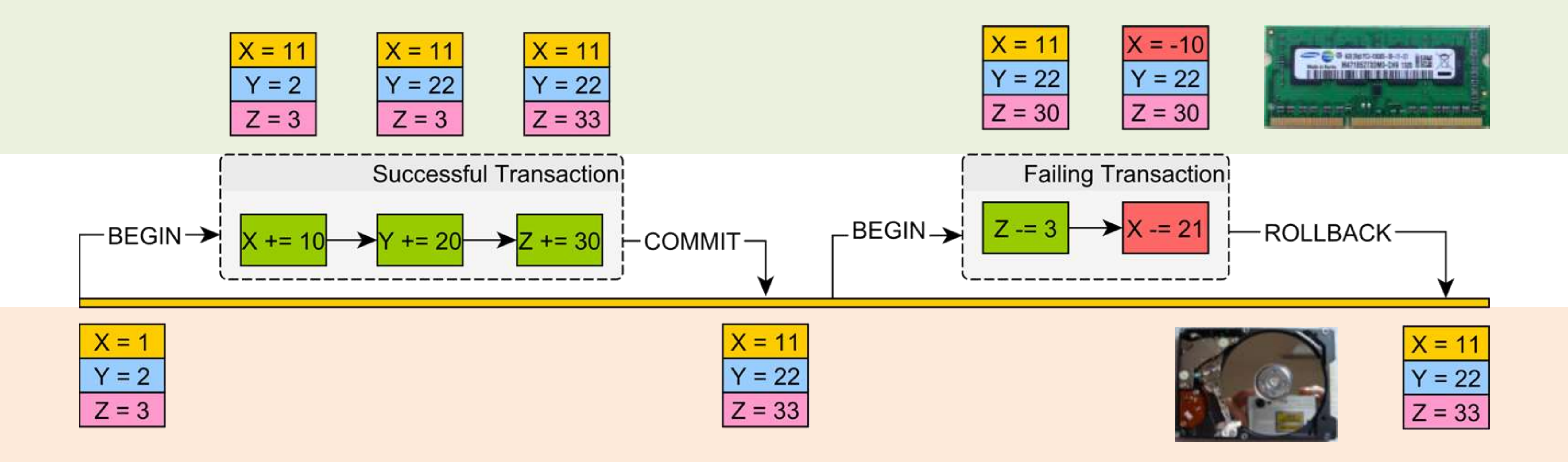
作要么

```
UPDATE checking_accounts
  SET balance = balance + 500
  WHERE account = 3208;
```

Increment Checking Account

```
INSERT INTO journal VALUES
  (journal_seq.NEXTVAL, '1B'
  3209, 3208, 500);
```

Record in Transaction Journal


X = 11
Y = 2
Z = 3
X = 11
Y = 22
Z = 3
X = 11
Y = 22
Z = 33
X = 11
Y = 22
Z = 30
X = -10
Y = 22
Z = 30
Successful Transaction
BEGIN
X += 10
Y += 20
Z += 30
COMMIT
Failing Transaction
BEGIN
Z -= 3
X -= 21
ROLLBACK
X = 1
Y = 2
Z = 3
X = 11
Y = 22
Z = 33
X = 11
Y = 22
Z = 33

ACID

Atomicity：原子性

- all or nothing
- 事务的操作要么全部执行，要么一个也不执行

Consistency：一致性

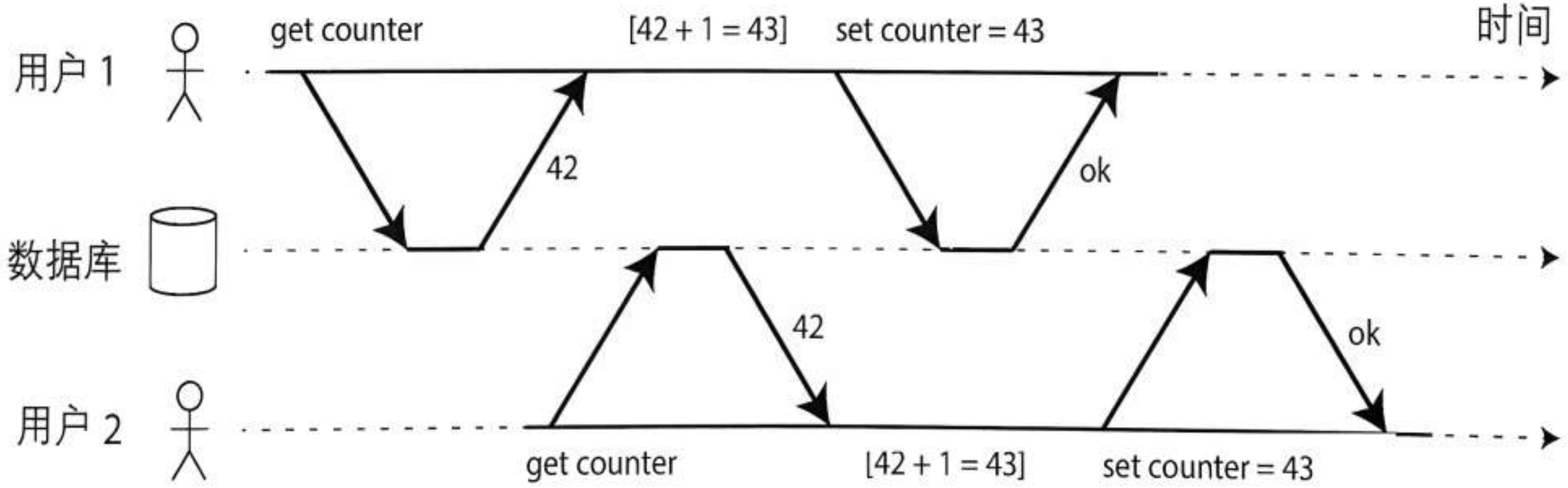
- it looks correct to me
- 如果事务程序正确且启动时处于一致状态，则事务结束时仍处于一致状态

Isolation：隔离性

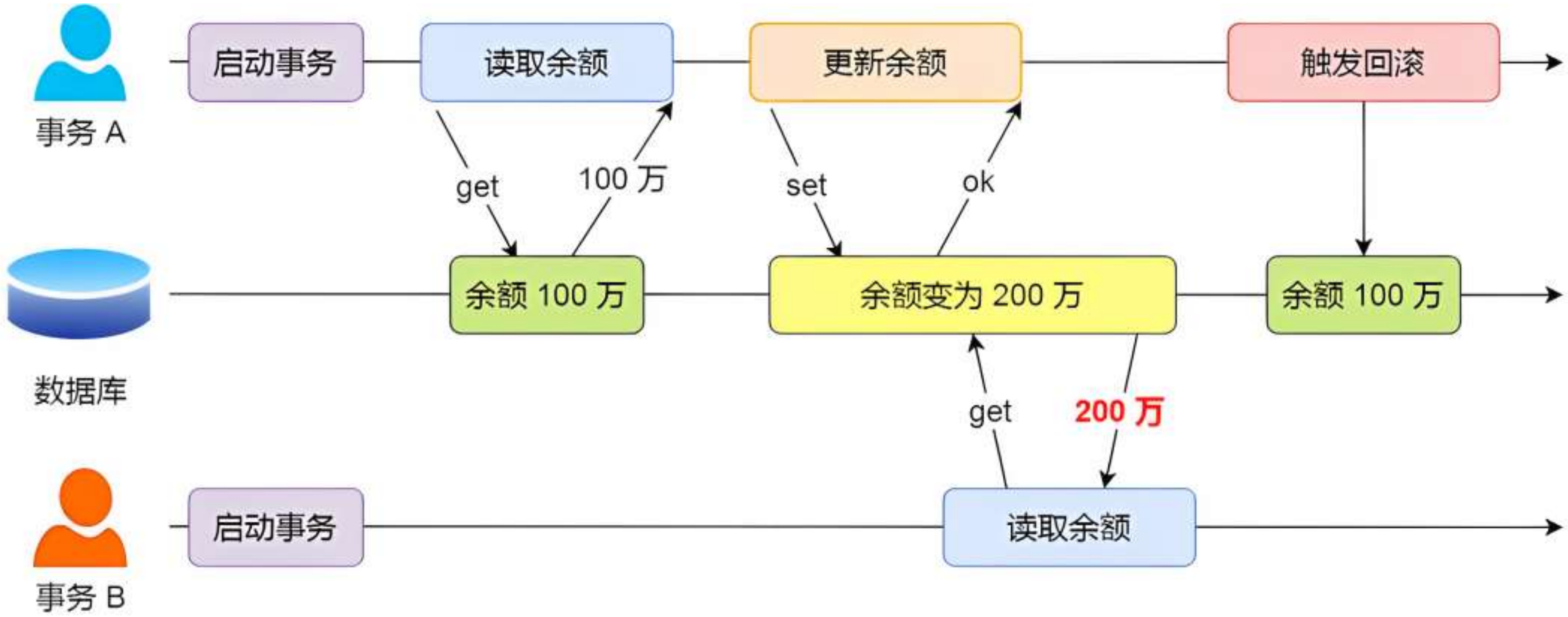
- as if alone
- 一个事务的执行不受其他事务的干扰

Durability：持久性

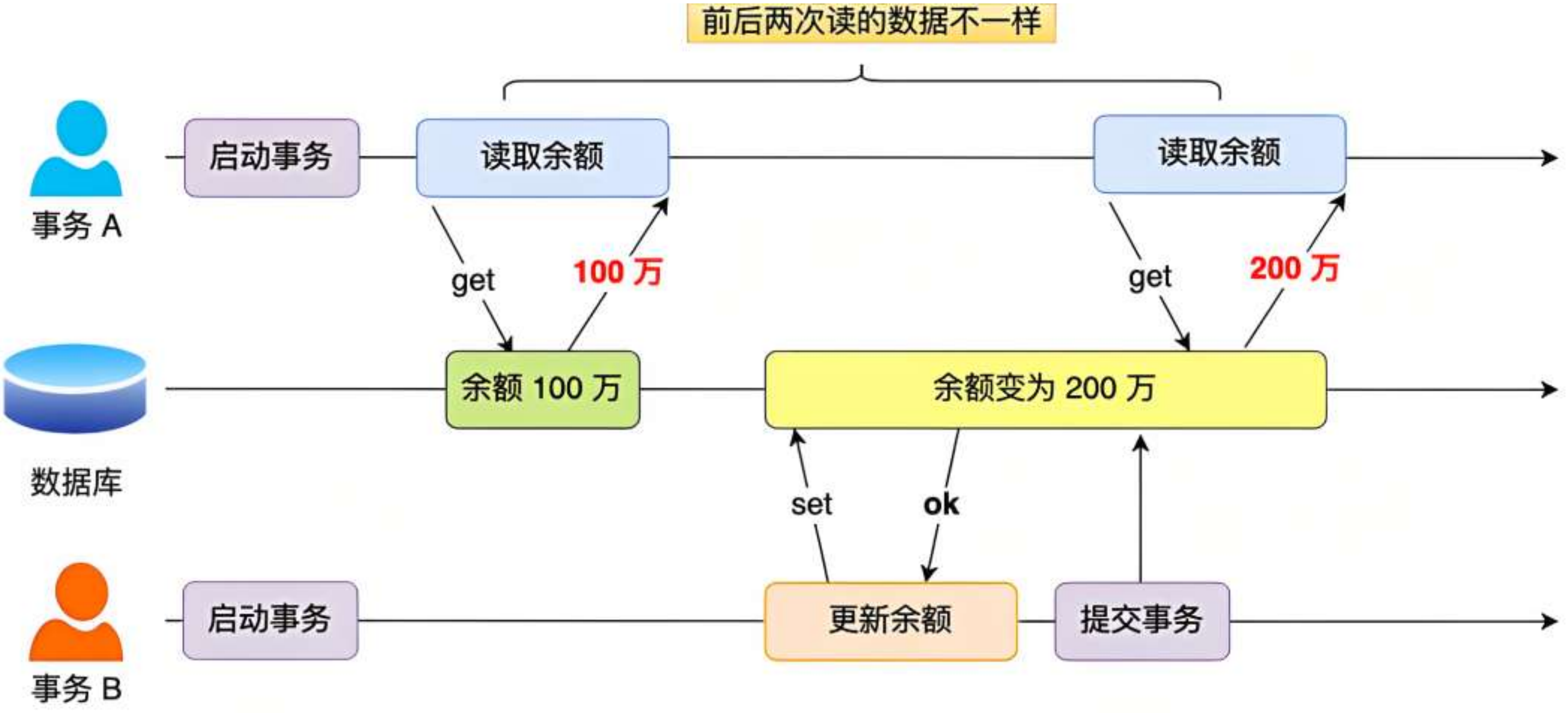
- survive failures
- 事务一旦提交，它对数据库的修改一定全部持久地写到数据库中



用户 1
get counter
[42 + 1 = 43]
set counter = 43
时间
42
ok
数据库
42
ok
用户 2
get counter
[42 + 1 = 43]
set counter = 43



事务 A
启动事务
读取余额
更新余额
触发回滚
get
100 万
set
ok
数据库
余额 100 万
余额变为 200 万
余额 100 万
get
200 万
事务 B
启动事务
读取余额


前后两次读的数据不一样
事务 A
启动事务
读取余额
读取余额
get
100 万
get
200 万
数据库
余额 100 万
余额变为 200 万
set
ok
事务 B
启动事务
更新余额
提交事务

数据管理

数据库系统

DBMS 的特色